



Aeroelastic modeling and analysis of honeycomb plates in high-speed airflow with acoustic load and general boundary conditions

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ABSTRACT

This paper investigates the modeling and aeroelastic properties of laminated plates with honeycomb core by a semi-analytical approach, where both acoustic load and general boundary conditions are considered. The equivalent material properties of the honeycomb structures are determined by the corrected Gibson's formula. The energy functional of the plate is derived based on the first-order shear deformation theory (FSDT) and the effect of aerodynamic pressure is taken into account by introducing the supersonic piston theory. A modified shear correction factor is adopted in this study to improve the accuracy of the method. The Hamilton's principle is employed to formulate the governing equations of the system, and the solutions of the problem with both classical and elastic boundary conditions are solved by a modified Ritz approach. The results obtained by the proposed method show good agreement with those by the finite element method and in open literature. Finally, parametric studies on the aeroelastic behaviors of the plate concerning the boundary conditions, layer thickness and honeycomb geometry are carried out.

1. Introduction

Sandwich plates with honeycomb core have been extensively used in aerospace, automobile, naval and many other fields where strength-to-weight ratio is a concern. It is known that structures in aerospace applications are usually exposed to severe aerodynamic and acoustic loads during cruise, which significantly affects their dynamic behaviors and may result in large structure response and panel flutter. When the flutter phenomenon occurs, the structure becomes unstable and keeps gathering energy from the airflow, which may cause fatigue failure and finally leads to catastrophic consequence. Thus, it is highly desirable to investigate the aeroelastic characteristics of honeycomb plates.

Panel flutter is a kind of self-excited vibration phenomenon where thin plates are subjected to supersonic flow directed parallel to their surfaces [1]. This phenomenon was first observed in 1940s [2] and gained much attention for causing structural failures in rockets and airplanes in the 1950s [3]. Since then, panel flutter has been extensively studied and a comprehensive review of the early studies on the panel flutter and aeroelastic stability of plates and shells was given by Dowell [4]. For panel flutter at supersonic speeds, both analytical and numerical methods are widely used to study its dynamic behaviors [5]. For several

special cases, the exact closed-form solutions were derived analytically by Sun and Xing [6]. For more general cases, semi-analytical methods are preferred due to their higher flexibility. One of the well-known methods in flutter analysis is the assumed mode method (AMM). By adopting AMM, Barai and Durvasula [7] analyzed the flutter of a 3D hybrid laminated panel and found that aerodynamic damping is important in determining the aeroelastic stability boundaries. Song and Li [8,9] studied the aero-elastic and aero-thermo-elastic flutter behaviors of composite laminated plates in supersonic flow and corresponding control strategies were proposed. Later, they successfully investigated the flutter properties of plates with classical and elastic boundary conditions using AMM [10]. However, it should be mentioned that despite its high convergence speed, the results may be unreliable for typical boundary conditions when using AMM [10]. To obtain more accurate results, the Ritz/Galerkin method is adopted to study the flutter characteristics of plates. Dowell [11,12] used Galerkin method for investigating the aero-elastic stability characteristics of a plate at supersonic speed, and he concluded that four to six modes should be used to obtain accurate results. Navazi and Haddadpour [13] studied the nonlinear aero-thermo-elastic behaviors of functionally graded plates in supersonic airflow. Shitov and Vedenev [14] investigated the structural

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stability of simply supported plates at low supersonic speeds by using the Bubnov–Galerkin method. For plates with aerodynamic load on both surfaces, the stability was studied by Wang et al [15] using the Galerkin method. By using the Ritz approach, Huang et al [16] studied the flutter characteristics of graphene nanoplatelet reinforced laminated composite quadrilateral plates with simply supported boundaries. Iterative enriched Ritz method was developed by Cunha-Filho et al [17] to deal with aeroviscoelastic panel flutter problems with both free and fixed boundaries. Muc and Flis [18] investigated the flutter characteristics of functionally graded porous plates with different boundary conditions using the Rayleigh-Ritz method. The same method is also adopted by Guimarães et al [19] to study the post-flutter behaviors of multibay composite panels and they found that multibay arrangements may affect the jump phenomenon in the Hopf bifurcation diagram. Although it is known that the traditional Ritz method has difficulty dealing with elastic boundaries, an improved form has been widely used to handle general boundaries by using the penalty method. By using the modified Ritz method with special treated admissible functions, Zhou et al [20] performed aero-thermo-elastic coupled flutter analysis on supersonic moderately thick orthotropic plates. It should be mentioned that this method is not limited to a single rectangular plate, but can deal with more complex structures as well [21,22]. The finite element method (FEM) is also a powerful tool in panel flutter analysis [23–25]. The FEM was first introduced by Olson [26] to deal with panel flutter problems, and can be used for many types of structures such as rectangular plates [27,28], curved plates [29–32] and stiffened plates [32–34]. By comparing with AMM, Song and Li [10] found that the results from the FEM are more precise than those of AMM.

As can be seen from the aforementioned literature, although various numerical methods have been developed to deal with panel flutter, a reliable plate model is always necessary for modeling panel structures. A comprehensive review on the modeling of plate structures can be found in Refs. [35–37]. Modeling of plate structures is actually a three-dimensional (3D) problem. However, the 3D elastic solution for laminated plates is usually very complex and difficult to obtain. To simplify the problem, researchers make some simplifications and reduce the problem from 3D to two-dimensional (2D) with reasonable accuracy. According to Carrera [38], the 2D assumptions of multilayered structures includes the Equivalent Single Layer (ESL) model and layer-wise (LW) model. According to the ESL theory, the displacement field of the plate is the expansion of its displacement parameters in the middle surface. Thus, the number of unknown variables is independent of the number of layers, which significantly simplifies the modeling and analysis of laminated plates. While in the LW theory, all layers are modeled as independent shell structures, thus the detailed response of individual layers can be obtained. Although adopting the LW theory usually leads to more accurate results, the solution procedure would become much more complex. Despite the differences between ESL and LW models, many researchers devoted themselves into finding unified solutions for laminated plates. The most well-known work is contributed by Carrera and known as Carrera Unified Formulation (CUF) [38,39], which provides a simple and elegant form for laminated structures. It is worthwhile to mention that CUF applies to not only laminated panels, but also beams and shells. Interested readers may refer to Hu et al [40] for a review and Refs. [41–45] for the latest developments.

For plates with honeycomb core, the honeycomb layer can be modeled as homogenous elastic solid with equivalent material properties. The most widely used approach for obtaining equivalent material properties of honeycomb structures was firstly introduced by Gibson and Ashby [46] and improved by Fu and Yin [47]. Since their method can be written in an analytical form, many researchers have adopted this approach in their studies [48–52]. Although this approach is widely accepted in many practical applications, it should be noted that the intricate microarchitecture-dependent mechanical characteristics of the honeycomb structures cannot be captured by this approach [53]. Recently, different higher-order continuum theories such as strain

gradient theory [53,54] and micropolar theory [55–57] have been developed to accurately model the deformation of lattice structures. However, as the constitutive parameters can hardly be written in an analytical form, numerical procedure is still necessary for determining these parameters when higher-order continuum theories are adopted.

To obtain a semi-analytical solution, the honeycomb core is modeled by equivalent material property in this paper. Thus, laminated plate theory can be adopted for the plate. There are mainly-three kinds of models widely used for laminated plates, namely, the classical plate theory (CPT), the first-order shell deformation theory (FSDT) and the higher-order shear deformation theory (HSDT). CPT is mainly used for plate structures when their thickness is considerably small compared with other dimensions. However, as CPT neglects the shear effect and rotary inertia, it becomes inaccurate when the thickness of the plate increases [58]. To overcome this limitation, FSDT is proposed and it has been proved to be accurate for moderately thick plates. Many researchers try to use FSDT for studying the dynamics of honeycomb plates. For example, Park and Mongeau [59] studied the vibration and sound radiation of a sandwich honeycomb panel using the FSDT. Duc et al [60–62] investigated the vibration and nonlinear response of honeycomb shells with negative Poisson's ratio. However, in benchmark results given by Madhukar and Singha [63], the natural frequencies predicted by FSDT and HSDT agree well only when the plate is very thin. Yu and Cleghorn [50] compared the results of the free vibration of symmetric honeycomb panels by using CPT, FSDT and HSDT, and pointed out that both CPT and FSDT are not adequate for the flexural vibration analysis of honeycomb panels. In their study, the natural frequencies of the honeycomb panel predicted by FSDT is much higher than those by HSDT unless the plate is extremely thin. Similar phenomenon is observed by Madhukar and Singha [64], which indicates FSDT cannot be directly used for moderately thick plates if accuracy is a concern. Thus, instead of FSDT, researches prefer HSDT when analyzing the dynamics of honeycomb plates. Zhang et al [65] used Reddy's third-order shear deformation theory to solve the free vibrations and nonlinear responses of a honeycomb sandwich plate. Kheirikhah and Akhavan-Rad [66] studied the free vibration of a composite-faced honeycomb plate using a high-order theory. Khare et al [67] presented a facet shell element based on higher-order theory for solving the dynamics of sandwich shells. Li and Zhu [68] introduced an iterative formulation based on Reddy's third-order plate theory and solved the free flexural vibration of multilayered honeycomb plates. Ansari et al [69] used HSDT along with von-Karman nonlinear kinematic relations to solve the nonlinear vibration of honeycomb plates with functionally graded carbon nanotube-reinforced composite face sheets. Nevertheless, although HSDT is more accurate for honeycomb plates, it leads to a much more complex formulation.

The reason FSDT is not appropriate for honeycomb plates is that the FSDT does not satisfy the zero shear stress condition on the upper and lower surfaces of a plate. For isotropic plates, a constant shear correction factor $5/6$ can be used to correct the shear stress. But for honeycomb plates, since the equivalent material properties of the honeycomb is quite soft compared with those of the face sheets, the FSDT would predict a much higher shear stress in the plate. Liu and Zhao [70] noticed that both FSDT and HSDT can obtain accurate results for honeycomb sandwich plates by simply neglecting the shear effects of the face sheets. Although their study is limited to sandwich plates where the shear modulus of face sheets is much higher than that of the core, and the thickness of the face sheets is much smaller than that of the core, it indicates that FSDT can still be accurate for honeycomb plates if the shear effect is carefully treated. However, obtaining accurate results for general honeycomb plates by FSDT still remains a challenge.

In this paper, a semi-analytical solution for the dynamics of a honeycomb plate is developed. The energy functional of the plate is derived based on FSDT and a modified shear correction factor is used to properly treat the shear effects. The aerodynamic pressure is modeled by the supersonic piston theory and the Hamilton's principle is employed to formulate the governing equations of the system. A modified Ritz

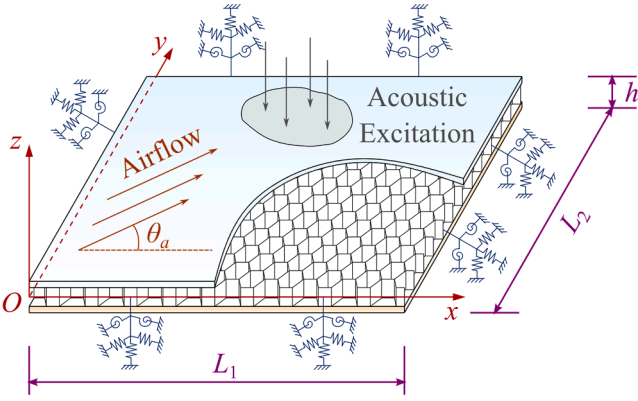


Fig. 1. Geometry and coordinate system of the laminated plate with honeycomb core.

approach is used to solve the free and forced vibration of the plate with general boundary conditions. Numerical verifications are conducted to show the accuracy of the proposed method. Parametric studies are also carried out to study the effect of some key parameters on the aerodynamic behaviors of the plate.

The remainder of the paper is arranged as follows. The governing equations of the laminated honeycomb plate with acoustic load in supersonic airflow and the corresponding solutions are presented in Section 2. Numerical studies and discussions are conducted in Section 3, and conclusions are drawn in Section 4.

2. Problem formulation

2.1. Model description

The structure of an acoustically loaded plate with honeycomb core layer and arbitrary boundary conditions is shown in Fig. 1. As shown in the figure, a Cartesian coordinate system is established to describe the geometry and deformation of the plate. The geometry of the plate can be defined by length L_1 , width L_2 and thickness h .

2.2. Kinematic relations and stress resultants

Based on the FSDT, the displacement field of the plate takes the following form [71]

$$\begin{aligned} u(x, y, z, t) &= u_0(x, y, t) + z\phi_x(x, y, t) \\ v(x, y, z, t) &= v_0(x, y, t) + z\phi_y(x, y, t) \\ w(x, y, z, t) &= w_0(x, y, t) \end{aligned} \quad (1)$$

where u_0 , v_0 and w_0 denote the displacements on the reference surface of the plate in the x , y and z directions, ϕ_x and ϕ_y denote the rotations of transverse normal with respect to y and x axis.

Thus, based on the linear elastic theory, the strain–displacement relations of the plate can be obtained [71]

$$\begin{aligned} \varepsilon_1 &= \varepsilon_1^0 + z \cdot \varepsilon_1^1 \\ \varepsilon_2 &= \varepsilon_2^0 + z \cdot \varepsilon_2^1 \\ \varepsilon_4 &= \varepsilon_4^0 \\ \varepsilon_5 &= \varepsilon_5^0 \\ \varepsilon_6 &= \varepsilon_6^0 + z \cdot \varepsilon_6^1 \end{aligned} \quad (2)$$

where

$$\begin{aligned} \varepsilon_1^0 &= \frac{\partial u_0}{\partial x}, \varepsilon_2^0 = \frac{\partial v_0}{\partial y}, \varepsilon_4^0 = \frac{\partial w_0}{\partial y} + \phi_y, \\ \varepsilon_5^0 &= \frac{\partial w_0}{\partial x} + \phi_x, \varepsilon_6^0 = \frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x} \\ \varepsilon_1^1 &= \frac{\partial \phi_x}{\partial x}, \varepsilon_2^1 = \frac{\partial \phi_y}{\partial y} \\ \varepsilon_6^1 &= \frac{\partial \phi_x}{\partial y} + \frac{\partial \phi_y}{\partial x} \end{aligned} \quad (3)$$

The normal and shear stresses of the laminated plate can be expressed by the generalized Hooke's law [71]

$$\begin{pmatrix} \sigma_1^k \\ \sigma_2^k \\ \sigma_4^k \\ \sigma_5^k \\ \sigma_6^k \end{pmatrix} = \begin{pmatrix} \bar{Q}_{11}^k & \bar{Q}_{12}^k & 0 & 0 & \bar{Q}_{16}^k \\ \bar{Q}_{12}^k & \bar{Q}_{22}^k & 0 & 0 & \bar{Q}_{26}^k \\ 0 & 0 & \bar{Q}_{44}^k & \bar{Q}_{45}^k & 0 \\ 0 & 0 & \bar{Q}_{45}^k & \bar{Q}_{55}^k & 0 \\ \bar{Q}_{16}^k & \bar{Q}_{26}^k & 0 & 0 & \bar{Q}_{66}^k \end{pmatrix} \begin{pmatrix} \varepsilon_1^k \\ \varepsilon_2^k \\ \varepsilon_4^k \\ \varepsilon_5^k \\ \varepsilon_6^k \end{pmatrix} \quad (4)$$

where σ_1^k and σ_2^k are the normal stresses of the k -th layer, σ_4^k , σ_5^k and σ_6^k are the corresponding shear stresses, and $\bar{Q}_{ij}^k$ denotes the elastic stiffness of the k -th layer. The expressions of $\bar{Q}_{ij}^k$ are [71]

$$\begin{aligned} \bar{Q}_{11}^k &= Q_{11}^k \cos^4 \theta_k + 2(Q_{12}^k + 2Q_{66}^k) \sin^2 \theta_k \cos^2 \theta_k + Q_{22}^k \sin^4 \theta_k \\ \bar{Q}_{12}^k &= (Q_{11}^k + Q_{22}^k - 4Q_{66}^k) \sin^2 \theta_k \cos^2 \theta_k + Q_{12}^k (\sin^4 \theta_k + \cos^4 \theta_k) \\ \bar{Q}_{22}^k &= Q_{11}^k \sin^4 \theta_k + 2(Q_{12}^k + 2Q_{66}^k) \sin^2 \theta_k \cos^2 \theta_k + Q_{22}^k \cos^4 \theta_k \\ \bar{Q}_{16}^k &= (Q_{11}^k - Q_{12}^k - 2Q_{66}^k) \sin \theta_k \cos^3 \theta_k + (Q_{12}^k - Q_{22}^k + 2Q_{66}^k) \sin^3 \theta_k \cos \theta_k \\ \bar{Q}_{26}^k &= (Q_{11}^k - Q_{12}^k - 2Q_{66}^k) \sin^3 \theta_k \cos \theta_k + (Q_{12}^k - Q_{22}^k + 2Q_{66}^k) \sin \theta_k \cos^3 \theta_k \\ \bar{Q}_{66}^k &= (Q_{11}^k + Q_{22}^k - 2Q_{12}^k - 2Q_{66}^k) \sin^2 \theta_k \cos^2 \theta_k + Q_{66}^k (\sin^4 \theta_k + \cos^4 \theta_k) \\ \bar{Q}_{44}^k &= Q_{44}^k \cos^2 \theta_k + Q_{55}^k \sin^2 \theta_k \\ \bar{Q}_{45}^k &= (Q_{55}^k - Q_{44}^k) \cos \theta_k \sin \theta_k \\ \bar{Q}_{55}^k &= Q_{55}^k \cos^2 \theta_k + Q_{44}^k \sin^2 \theta_k \end{aligned} \quad (5)$$

where θ_k denotes the orientation angle between the principal direction of the k -th layer and the x -axis, and the elastic stiffness coefficients $\bar{Q}_{ij}^k$ are given by [71]

$$\begin{aligned} Q_{11}^k &= \frac{E_1^k}{1 - \nu_{12}^k \nu_{21}^k} \\ Q_{12}^k &= \frac{\nu_{21}^k E_1^k}{1 - \nu_{12}^k \nu_{21}^k} \\ Q_{22}^k &= \frac{E_2^k}{1 - \nu_{12}^k \nu_{21}^k} \\ Q_{44}^k &= G_{23}^k, Q_{55}^k = G_{13}^k, Q_{66}^k = G_{12}^k \end{aligned} \quad (6)$$

where E_1^k , E_2^k , G_{23}^k , G_{13}^k , G_{12}^k and ν_{12}^k are the Young's moduli, the shear moduli and the Poisson's ratios of the k -th layer, respectively. It should be noted that the Poisson's ratios satisfy the following equation

$$\frac{\nu_{12}^k}{E_1^k} = \frac{\nu_{21}^k}{E_2^k} \quad (7)$$

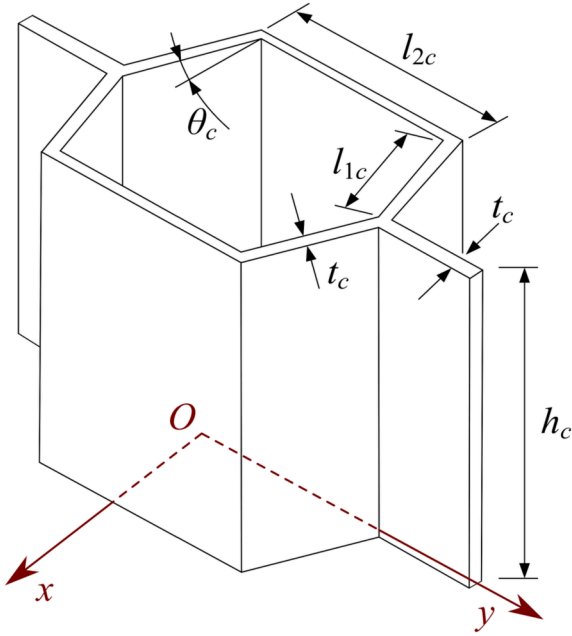


Fig. 2. Geometry of the hexagonal honeycomb cell.

For the honeycomb layer, the material properties cannot be used directly and equivalent elastic properties should be used instead. The expressions of the equivalent material properties depend on both the material and the shape of the honeycomb structure. Hexagonal honeycomb is the most widely used in practice and the geometry of its cell is shown in Fig. 2. The honeycomb structure is made of isotropic material with Young's modulus E_c , shear modulus G_c and density ρ_c . Thus, the equivalent material properties can be calculated by the corrected Gibson's formula [46,47]

$$\begin{aligned}
 E_1^k &= E_c \eta_2^3 \frac{1}{(\tan^2 \theta_c + \eta_2^2)(\eta_1 + \sin \theta_c) \cos \theta_c} \\
 E_2^k &= E_c \eta_2^3 \frac{\eta_1 + \sin \theta_c}{(\cos^2 \theta_c + (\eta_1 + \sin^2 \theta_c) \eta_2^2) \cos \theta_c} \\
 \nu_{12}^k &= \frac{(1 - \eta_2^2) \sin \theta_c}{(\tan^2 \theta_c + \eta_2^2)(\eta_1 + \sin \theta_c)} \\
 \nu_{21}^k &= \frac{(1 - \eta_2^2)(\eta_1 + \sin \theta_c) \sin \theta_c}{(1 + (\eta_1 \sec^2 \theta_c + \tan^2 \theta_c) \eta_2^2) \cos^2 \theta_c} \\
 G_{23}^k &= G_c \frac{\eta_2}{2 \cos \theta_c} \left(\frac{\eta_1 + \sin \theta_c}{1 + 2\eta_1} + \frac{\eta_1 + 2 \sin^2 \theta_c}{2(\eta_1 + \sin \theta_c)} \right) \\
 G_{13}^k &= G_c \frac{\eta_2 \cos \theta_c}{\eta_1 + \sin \theta_c} \\
 G_{12}^k &= E_c \eta_2^3 \frac{\eta_1 + \sin \theta_c}{\eta_1^2 (1 + 2\eta_1) \cos \theta_c} \\
 \rho^k &= \rho_c \eta_2 \frac{1 + \eta_1}{(\eta_1 + \sin \theta_c) \cos \theta_c}
 \end{aligned} \quad (8)$$

where θ_c is the cell angle, $\eta_1 = l_{2c}/l_{1c}$ is the aspect ratio and $\eta_2 = t_c/l_{1c}$ is the non-dimensional thickness of cell walls. It should be mentioned that the shear modulus G_{23}^k is the average of the upper and lower bounds given by Ref. [46].

The in-plane force resultants $(N_{11}, N_{22}, N_{12})^T$ and moment resultants $(M_{11}, M_{22}, M_{12})^T$ can thus be obtained by integrating the normal and

shear stresses along the thickness direction of the plate

$$\begin{pmatrix} N_{11} \\ N_{22} \\ N_{12} \end{pmatrix} = \sum_k \int_{z_k} \begin{pmatrix} \sigma_1^k \\ \sigma_2^k \\ \sigma_6^k \end{pmatrix} dz, \quad \begin{pmatrix} M_{11} \\ M_{22} \\ M_{12} \end{pmatrix} = \sum_k \int_{z_k} \begin{pmatrix} \sigma_1^k \\ \sigma_2^k \\ \sigma_6^k \end{pmatrix} z dz \quad (9)$$

According to FSDT, the out-of-plane transverse shear stresses σ_4 and σ_5 are assumed to be constant throughout the thickness direction of the plate, which would lead to a decrease of accuracy. This problem is solved by multiplying the transverse stress by a shear correction factor. For homogeneous plate, the shear correction factor κ is typically taken as 5/6 as given by Ref. [37,63]. Nevertheless, for typical sandwich structures with soft cores, directly using $\kappa = 5/6$ results in a relatively large error [50,63,64]. This is due to the fact that considering the zero shear stress condition, the out-of-plane shear stresses in the face layers are actually small and letting $\kappa = 5/6$ over estimates the shear effect, which finally results in a much higher natural frequency. To overcome the problem, a practical workaround is to neglect the shear effects of the face sheets when they are relatively thin [70]. Another solution to the problem is using a modified version of HSDT instead of FSDT to get more accurate solutions for plates with a honeycomb cores [68]. However, using HSDT leads to a much more complex formulation than using FSDT.

To overcome the problem, a more accurate shear correction factor is introduced to eliminate the inaccuracy induced by the shear effect of the face layers. Thus, a simpler and more accurate solution to the vibration of the plate can be obtained by FSDT.

By introducing the shear correction factor, shear force resultants can be expressed as

$$\begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = \begin{pmatrix} \kappa_1 & 0 \\ 0 & \kappa_2 \end{pmatrix} \sum_k \int_{z_k} \begin{pmatrix} \sigma_4 \\ \sigma_5 \end{pmatrix} dz \quad (10)$$

in which κ_1 and κ_2 are shear correction factors given by [72]

$$\kappa_\alpha = \frac{R_\alpha^2}{d_\alpha J_\alpha}, \quad \alpha = 1, 2 \quad (11)$$

where

$$R_\alpha = \sum_k \int_{z_k} \bar{Q}_{\alpha\alpha} (z - z_{n\alpha})^2 dz$$

$$d_\alpha = \sum_k \int_{z_k} \bar{Q}_{(6-\alpha)3} dz$$

$$I_\alpha = \sum_k \int_{z_k} \frac{g_\alpha^2(z)}{\bar{Q}_{(6-\alpha)3}} dz$$

$$g_\alpha(z) = - \int_{z_0}^z \bar{Q}_{\alpha\alpha} (z - z_{n\alpha}) dz$$

$$z_{n\alpha} = \frac{\sum_k \int_{z_k} \bar{Q}_{\alpha\alpha} z dz}{\sum_k \int_{z_k} \bar{Q}_{\alpha\alpha} dz} \quad (12)$$

It is not difficult to find out that for homogeneous plate, $\kappa_1 = \kappa_2 = 5/6$, which falls back to the traditional case.

The aerodynamic pressure induced by supersonic airflow can be estimated by the first-order piston theory [20]

$$\Delta p = \frac{\rho_\infty U_\infty^2}{\sqrt{M_\infty^2 - 1}} \left(\frac{M_\infty^2 - 2}{M_\infty^2 - 1} \frac{1}{U_\infty} \frac{\partial w}{\partial t} + \frac{\partial w}{\partial x} \cos \theta_a + \frac{\partial w}{\partial y} \sin \theta_a \right) \quad (13)$$

where ρ_∞ , U_∞ and M_∞ denote the air density, velocity and Mach number of the free stream, respectively.

For the sake of brevity in the following analysis, the non-dimensional aerodynamic pressure λ is adopted [20]

$$\lambda = \frac{\rho_{\infty} U_{\infty}^2 L_1^3}{D_1^1 \sqrt{M_{\infty}^2 - 1}} \quad (14)$$

In the expression, $D_1^1 = E_1^1 h^3 / [12(1 - \nu_{12}^1 \nu_{21}^1)]$, where E_1^1 , ν_{12}^1 and ν_{21}^1 are the elastic parameters of the first layer of the plate.

Substituting Eq. (2) and Eq. (4) into Eq. (9) leads to

$$\begin{pmatrix} N_{11} \\ N_{22} \\ N_{12} \\ M_{11} \\ M_{22} \\ M_{12} \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ B_{11} & B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{pmatrix} \begin{pmatrix} \varepsilon_1^0 \\ \varepsilon_2^0 \\ \varepsilon_6^0 \\ \varepsilon_1^1 \\ \varepsilon_2^1 \\ \varepsilon_6^1 \end{pmatrix} \quad (15)$$

where

$$[A_{ij} \quad B_{ij} \quad D_{ij}] = \sum_k \int_{z_k} \bar{Q}_{ij}^k [1 \quad z \quad z^2] dz \quad (16)$$

Similarly, substituting Eq. (2) and Eq. (4) into Eq. (10) leads to

$$\begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = \begin{pmatrix} \kappa_1 & 0 \\ 0 & \kappa_2 \end{pmatrix} \begin{pmatrix} A_{44} & A_{45} \\ A_{45} & A_{55} \end{pmatrix} \begin{pmatrix} \varepsilon_4^0 \\ \varepsilon_5^0 \end{pmatrix} \quad (17)$$

2.3. Energy expressions

The virtual strain energy of the laminated plate is expressed as [71]

$$\delta U_e = \iint_{\Omega} (N_{11} \delta \varepsilon_1^0 + N_{22} \delta \varepsilon_2^0 + N_{66} \delta \varepsilon_6^0 + M_{11} \delta \varepsilon_1^1 + M_{22} \delta \varepsilon_2^1 + M_{66} \delta \varepsilon_6^1 + Q_1 \delta \varepsilon_4^0 + Q_2 \delta \varepsilon_5^0) d\Omega \quad (18)$$

The virtual elastic energy stored in the boundary springs are given by [73]

$$\delta U_{BC} = \iint_{\Omega} (k_1 u_0 \delta u_0 + k_2 v_0 \delta v_0 + k_3 w_0 \delta w_0 + k_4 \phi_x \delta \phi_x + k_5 \phi_y \delta \phi_y) d\Omega \quad (19)$$

where k_1, k_2, k_3, k_4, k_5 are the stiffness coefficients for the five degrees of freedom (DOF), respectively. Eq. (19) integrates the virtual energy over the whole plate, which is the virtual energy for elastic foundations [74]. To account for the virtual energy of the boundary conditions of the four edges of the plate, the integral should be evaluated along the four edges.

$$\delta U_{BC} = \int_{\partial\Omega} (k_1 u_0 \delta u_0 + k_2 v_0 \delta v_0 + k_3 w_0 \delta w_0 + k_4 \phi_x \delta \phi_x + k_5 \phi_y \delta \phi_y) dl \quad (20)$$

And for isolated spring located at (x_0, y_0) , the virtual energy term for boundary conditions is

$$\delta U_{BC} = (k_1 u_0 \delta u_0 + k_2 v_0 \delta v_0 + k_3 w_0 \delta w_0 + k_4 \phi_x \delta \phi_x + k_5 \phi_y \delta \phi_y) |_{x=x_0, y=y_0} \quad (21)$$

The virtual kinetic energy of the plate is [71]

$$\begin{aligned} \delta K = & \iint_{\Omega} \left(I_0 (\dot{u}_0 \delta \dot{u}_0 + \dot{v}_0 \delta \dot{v}_0 + \dot{w}_0 \delta \dot{w}_0) + I_1 (\dot{u}_0 \delta \dot{\phi}_x + \dot{\phi}_x \delta \dot{u}_0 + \dot{v}_0 \delta \dot{\phi}_y + \dot{\phi}_y \delta \dot{v}_0) \right. \\ & \left. + I_2 (\dot{\phi}_x \delta \dot{\phi}_x + \dot{\phi}_y \delta \dot{\phi}_y) \right) d\Omega \end{aligned} \quad (22)$$

where I_0, I_1 and I_2 are the mass moments of inertia, and can be expressed as

$$[I_0 \quad I_1 \quad I_2] = \sum_k \int_{z_k} \rho_k [1 \quad z \quad z^2] dz \quad (23)$$

Acoustic load can be treated as stochastic excitation $p_s(x, y, t)$ perpendicularly exerted on the plate surface. Thus, the virtual work of the acoustic load can be written as [75]

$$\delta W_s = \int_{\Omega} p_s \delta w d\Omega \quad (24)$$

And the virtual work done by the airflow is [75]

$$\delta W_a = \int_{\Omega} \Delta p \delta w d\Omega \quad (25)$$

2.4. Solutions

By adopting the Hamilton's principle, the following equation can be obtained

$$\int_{t_0}^{t_1} (\delta K - \delta U_e - \delta U_{BC} + \delta W_s + \delta W_a) dt = 0 \quad (26)$$

Substituting the detailed expressions of the energy functionals into Eq. (26), the dynamics of the plate can be expressed by an equation concerning u_0, v_0, w_0, ϕ_x and ϕ_y . The explicit forms of the governing equations and the corresponding boundary conditions can also be obtained by integrating Eq. (26) by parts and are given in Appendix A.

In order to obtain an accurate solution for the dynamics of the plate, a set of admissible functions should be selected to express the displacement of the plate

$$\begin{aligned} u_0(x, y, t) &= \sum_{m=0}^M \sum_{n=0}^N U_{mn}(t) \varphi_m\left(\frac{x}{L_1}\right) \varphi_n\left(\frac{y}{L_2}\right) \\ v_0(x, y, t) &= \sum_{m=0}^M \sum_{n=0}^N V_{mn}(t) \varphi_m\left(\frac{x}{L_1}\right) \varphi_n\left(\frac{y}{L_2}\right) \\ w_0(x, y, t) &= \sum_{m=0}^M \sum_{n=0}^N W_{mn}(t) \varphi_m\left(\frac{x}{L_1}\right) \varphi_n\left(\frac{y}{L_2}\right) \\ \phi_1(x, y, t) &= \sum_{m=0}^M \sum_{n=0}^N \Phi_{1mn}(t) \varphi_m\left(\frac{x}{L_1}\right) \varphi_n\left(\frac{y}{L_2}\right) \\ \phi_2(x, y, t) &= \sum_{m=0}^M \sum_{n=0}^N \Phi_{2mn}(t) \varphi_m\left(\frac{x}{L_1}\right) \varphi_n\left(\frac{y}{L_2}\right) \end{aligned} \quad (27)$$

where $U_{mn}(t), V_{mn}(t), W_{mn}(t), \Phi_{1mn}(t)$ and $\Phi_{2mn}(t)$ are the generalized coordinates, $\varphi_k(x)$ is the corresponding admissible function. The admissible functions need to be linearly independent when using the Ritz approach. In many cases, Fourier series is the most widely used admissible function set due to its orthogonality. However, it is reported that using the Fourier series may lead to a poor convergence behavior [76]. Thus, in order to improve convergence, the Legendre polynomials are used instead in this study

$$\varphi_k(x) = \frac{1}{k!} \frac{d^k}{dx^k} \left((x^2 - x)^k \right) \quad (28)$$

It should be noted that the Legendre polynomials have infinite terms

Table 1
Equivalent material properties and shear correction factors.

E_1^2	E_2^2	G_{23}^2	G_{13}^2	G_{12}^2	ν_{12}	ν_{21}	κ_1	κ_1
47.26 MPa	47.55 MPa	1.013GPa	1.013GPa	1.197GPa	0.9825	0.9882	0.1688	0.1687

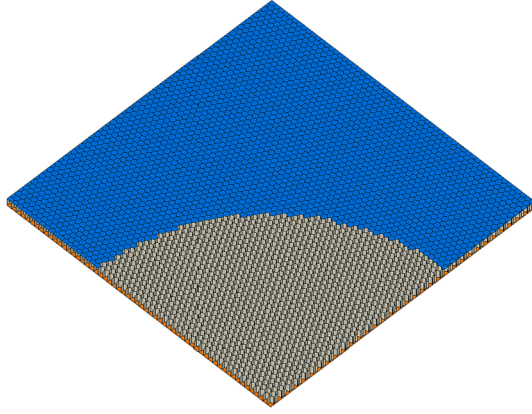


Fig. 3. Finite element model of the shell.

as expressed in Eq. (28), and should be truncated in practical calculation. In this case, the orders of the Legendre polynomials are uniformly truncated to M and N , respectively.

Eq. (27) can be written in a more compact form by using vector products

$$\begin{aligned}
 u_0(x, y, t) &= \mathbf{U}_0^T(t) \Psi(x, y) \\
 v_0(x, y, t) &= \mathbf{V}_0^T(t) \Psi(x, y) \\
 w_0(x, y, t) &= \mathbf{W}_0^T(t) \Psi(x, y) \\
 \phi_1(x, y, t) &= \Phi_1^T(t) \Psi(x, y) \\
 \phi_2(x, y, t) &= \Phi_2^T(t) \Psi(x, y)
 \end{aligned} \quad (29)$$

where $\mathbf{U}_0(t)$, $\mathbf{V}_0(t)$, $\mathbf{W}_0(t)$, $\Phi_1(t)$, $\Phi_2(t)$ and $\Psi(x, y)$ are column vectors of length $M \times N$ containing $U_{mn}(t)$, $V_{mn}(t)$, $W_{mn}(t)$, $\Phi_{1mn}(t)$, $\Phi_{2mn}(t)$ and $\varphi_m(x/L_1) \cdot \varphi_n(y/L_2)$, respectively.

Substituting Eq. (29) into Eq. (26), the equations of motion of the plate can be obtained in the matrix form

$$\mathbf{M}_s \ddot{\mathbf{q}} + \mathbf{C}_d \dot{\mathbf{q}} + (\mathbf{K}_s + \mathbf{K}_{BC} + \mathbf{K}_a) \mathbf{q} = \mathbf{F}_a \quad (30)$$

where $\mathbf{M}_s$ is the mass matrix, $\mathbf{C}_d$ is the damping matrix, $\mathbf{K}_s$, $\mathbf{K}_{BC}$ and $\mathbf{K}_a$

are the stiffness matrixes for the plate structure, boundary conditions and aerodynamic pressure respectively, $\mathbf{F}_a$ is the force vector and $\mathbf{q} = (\mathbf{U}_0^T, \mathbf{V}_0^T, \mathbf{W}_0^T, \Phi_1^T, \Phi_2^T)$ is the generalized displacement vector to be solved. The detailed expressions of the matrixes are given in Appendix B.

Thus, the equations of motion of the plate in the frequency domain can be written in the following form

$$-\omega^2 \mathbf{M}_s \bar{\mathbf{q}} + j\omega \mathbf{C}_d \bar{\mathbf{q}} + (\mathbf{K}_s + \mathbf{K}_{BC} + \mathbf{K}_a) \bar{\mathbf{q}} = \bar{\mathbf{F}}_a \quad (31)$$

where $j = \sqrt{-1}$ is the imaginary unit.

Solving the free vibration problem requires calculating the non-trivial solution of Eq. (30) when the external force vector $\mathbf{F}_a = 0$. Letting $\mathbf{q}_0 = \mathbf{q}$, $\dot{\mathbf{q}}_1 = \dot{\mathbf{q}}$, the following matrix equation can be obtained

$$\begin{pmatrix} \dot{\mathbf{q}}_0 \\ \mathbf{q}_1 \end{pmatrix} = \begin{pmatrix} 0 & \mathbf{I} \\ -\mathbf{M}_s^{-1}(\mathbf{K}_s + \mathbf{K}_{BC} + \mathbf{K}_a) & -\mathbf{M}_s^{-1} \mathbf{C}_d \end{pmatrix} \begin{pmatrix} \mathbf{q}_0 \\ \mathbf{q}_1 \end{pmatrix} \quad (32)$$

Eq. (32) transfers Eq. (30) into the standard form of dynamical systems. Thus, the natural frequencies and mode shapes of the plate can be obtained by solving the eigenvalue and eigenvectors of Eq. (32).

3. Results and discussion

3.1. Convergence and validation study

Based on the proposed formulations, a corresponding computer program is developed using Python. The convergence study concerning the truncation of the admissible functions is discussed first. In this study, a laminated sandwich plate with honeycomb core is considered. The plate is made of aluminum, and the material properties are: $E=70$ GPa, $\nu=0.33$ and $\rho=2710$ kg/m³. The length L_1 and width L_2 of the plate are both 0.3 m and the thicknesses of the two face layers are $h_1 = h_3 = 1$ mm, while the thickness of the honeycomb core is $h_2 = 8$ mm. The geometric parameters of the honeycombs are: $t_c = 0.2$ mm, $l_{1c} = l_{2c} = 3$ mm, $\theta_c = 30^\circ$, and the orientation angle of the honeycomb layer is $\theta_2 = 0^\circ$. All four edges of the plate are clamped in this case. By adopting Eq. (8) and (11), the equivalent material properties of the honeycomb core and the shear correction factors can be calculated and are listed in Table 1. It can be found from the table that the honeycomb layer is almost isotropic in this case.

In order to find a suitable truncation number for the Legendre polynomials, the first ten natural frequencies of the plate with different

Table 2
Natural frequencies of the plate with different numbers of admissible functions.

Mode	Natural frequencies (Hz)						FEM	Error*
	$M=N=10$	$M=N=11$	$M=N=12$	$M=N=13$	$M=N=14$	$M=N=15$		
1	1172.2	1172.2	1172.2	1172.2	1172.2	1172.2	1217.9	-3.8 %
2	2224.7	2224.7	2224.7	2224.7	2224.7	2224.7	2312.6	-3.8 %
3	2224.8	2224.8	2224.7	2224.7	2224.7	2224.7	2313.8	-3.8 %
4	3110.6	3110.5	3110.5	3110.5	3110.5	3110.5	3236.1	-3.9 %
5	3655.3	3654.6	3654.6	3654.6	3654.6	3654.6	3803.4	-3.9 %
6	3689.5	3688.7	3688.7	3688.7	3688.7	3688.7	3838.7	-3.9 %
7	4415.3	4414.8	4414.7	4414.7	4414.7	4414.7	4594.7	-3.9 %
8	4415.4	4414.9	4414.8	4414.8	4414.8	4414.8	4597.6	-4.0 %
9	5366.9	5361.2	5357.5	5357.3	5357.3	5357.3	5575.4	-3.9 %
10	5367.1	5361.4	5357.6	5357.5	5357.5	5357.5	5584.7	-4.1 %
Time (s)	0.092	0.125	0.14	0.166	0.223	0.304	1898	/

* Denoting the frequencies by the proposed method with $M = N = 15$ as f_e and those by FEM as f_a , the error is defined as $\text{Error} = (f_e - f_a)/f_a \times 100\%$.

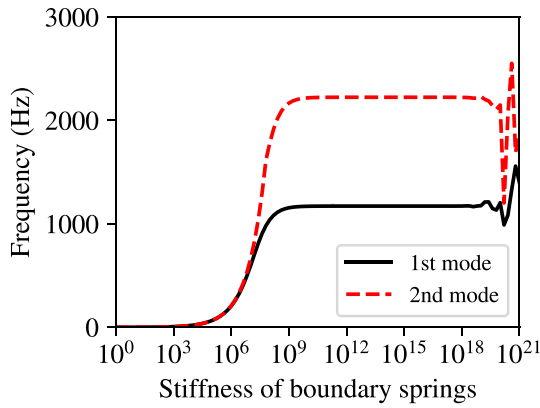


Fig. 4. The variation of the first two natural frequencies with different stiffness values of the boundary springs.

Table 3

Geometric and material properties of the plate given by Ref. [10].

L_1 (m)	L_2 (m)	h (m)	E (GPa)	ρ (kg/m ³)	ν
0.1	0.1	0.001	210	7930	0.33

numbers of admissible functions are calculated using the proposed method. To verify the results, a corresponding finite element model is also established by using commercial software ABAQUS as shown in Fig. 3. In the finite element model, two face layers of the plate and the internal structures of the honeycomb core are built by shell elements and a total of 422,013 elements are used.

Table 2 lists the natural frequencies of the plate obtained by the proposed method and the FEM results along with the computational time. It shows that the results of the proposed method match well with

those of the FEM, and the convergence shows that $M = N = 15$ is sufficient for the analysis. An interesting discovery is that the natural frequencies obtained by the proposed method is about 4 % lower than those by FEM. This finding agrees with the results given by Karttunen et al [55], which also shows that classical theory underestimates the natural frequencies of lattice panels. This is due to the fact that classical theory ignores the microstructure-dependent behaviors of lattice structures such as antisymmetric shear effect and local bending deformations [55]. To overcome these limitations, the strain gradient theory [53,54] and micropolar theory [55–57] can be adopted to get a more accurate solution.

In the present study, the boundary conditions are implemented by adding their potential energy term to the total energy functional. Thus, it is possible to take general boundary conditions into consideration by setting proper boundary stiffness parameters k_i . For example, the clamped boundary condition can be obtained by setting all boundary stiffness parameters to a sufficient large value while the free boundary condition can be obtained by setting them to 0. The effect of the stiffness of the boundary springs on the natural frequencies of the plate is shown in Fig. 4. In this study, the values of the boundary stiffnesses are changed simultaneously from 10^0 to 10^{21} , and the first two natural frequencies of the plate are calculated by the proposed method. It is clearly observed that the natural frequencies of the plate remain constant when the values of stiffness exceed 10^{10} . However, when the values of stiffness continue increasing and reach 10^{19} , the results become unstable due to the ill-conditioning problem [77]. Thus, to get correct results, a proper stiffness value should be chosen for fixed boundaries. Although the most appropriate value for fixed boundaries depends on the specific problem and the computational environment, it is usually acceptable to choose values between 10^{12} to 10^{15} for this kind of problem.

Next, another example is given to validate the natural frequencies of plates under aerodynamic load by comparing the results of the present study with those in open literature. In this example, an isotropic panel is considered and its geometric and material properties are given in

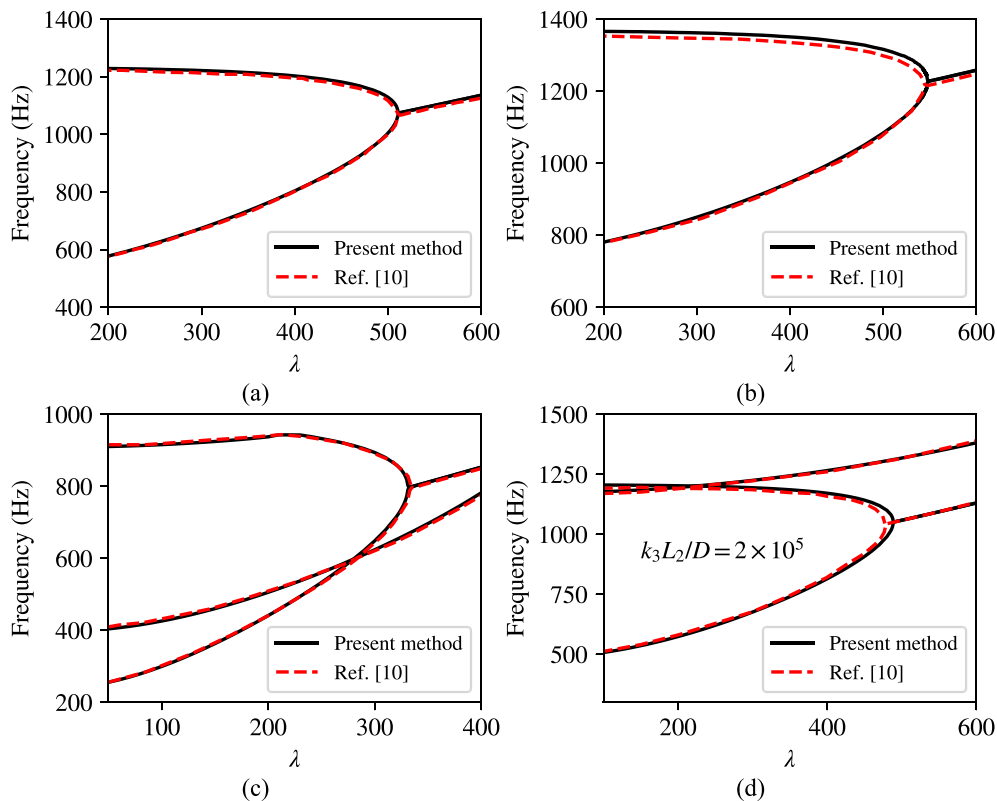


Fig. 5. Comparisons of natural frequencies of the plate with different aerodynamic pressure: (a) S-S-S-S, (b) S-C-S-C, (c) S-F-S-F, (d) S-Se-S-Se.

Table 4

Geometric and material properties of the plate given by Ref. [78].

L_1 (m)	L_2 (m)	h (m)	E (GPa)	ρ (kg/m ³)	ν
0.4	0.2	0.002	70	2700	0.33

Table 3. Four cases with different boundary conditions are considered. As is shown in Fig. 5, the first three cases study the natural frequencies of the plate with classical boundary conditions, where “S”, “C” and “F” denote simply supported, clamped and free boundary conditions, respectively. In the last case, two edges are clamped and the other two edges are elastically supported in the z direction, which is denoted by “Se”. The natural frequencies of the four cases are compared with those in Ref. [10] and the results are shown in Fig. 5, where the non-dimensional parameter D is defined as $D = Eh^3/(12(1 - \nu^2))$. From the figures, excellent match can be observed between the obtained results and those from literature, which indicates the proposed method can accurately evaluate the flutter behaviors of plates with various boundary

conditions.

Finally, the results for the stochastic vibration of the plate induced by stationary random load are compared with the benchmark results given by Ref. [78]. The geometric and material properties of the plate are given in Table 4. The plate is subjected to a band-limited white noise base excitation of acceleration with PSD = 0.5 g²/Hz within [20, 2000] Hz. In the calculation, the modal damping is set to 0.05 and the results of the acceleration response PSD at the central point are compared with those from Ref. [78]. Fig. 6 shows the comparison of the PSD functions with different boundary conditions and a good agreement can be observed between the obtained results and those from literature.

3.2. Numerical study and discussion

With the proposed method and computer program verified above, numerical studies are further carried out to study the flutter characteristics and the response of sandwich plates with honeycomb core under acoustic excitation. The plate is the same as that used for convergence

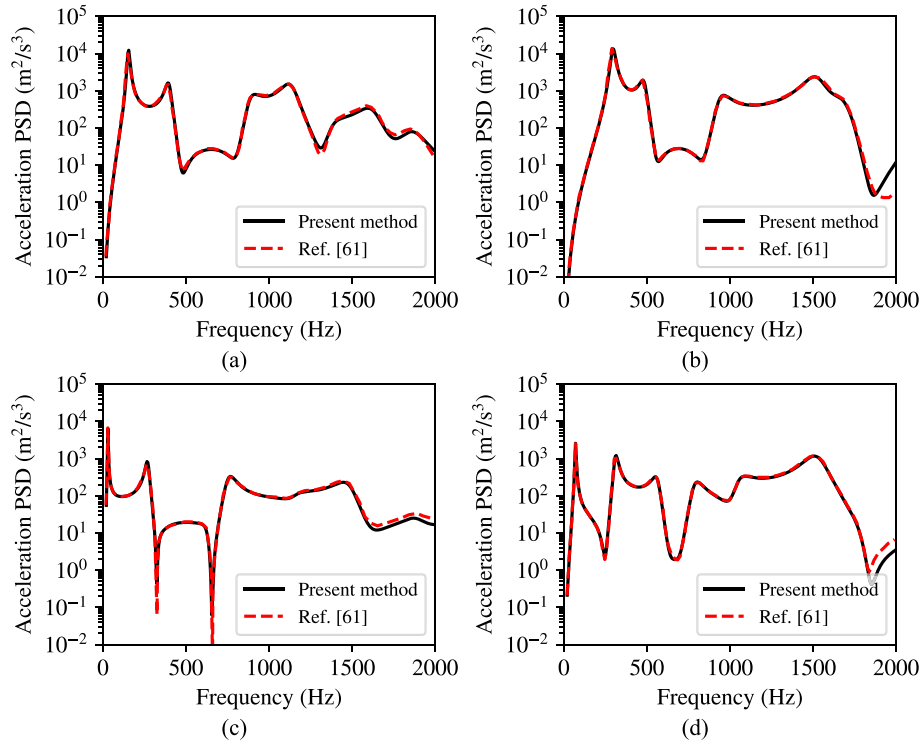


Fig. 6. Comparisons of acceleration response PSD with different boundary conditions: (a) S-S-S-S, (b) S-C-S-C, (c) S-F-S-F, (d) S-C-S-F.

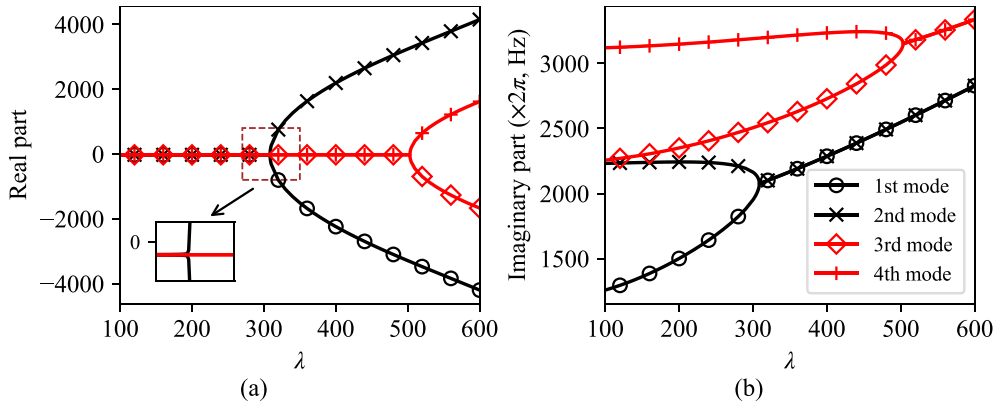


Fig. 7. Variations of the eigenvalues with aerodynamic pressure: (a) real part, (b) imaginary part.

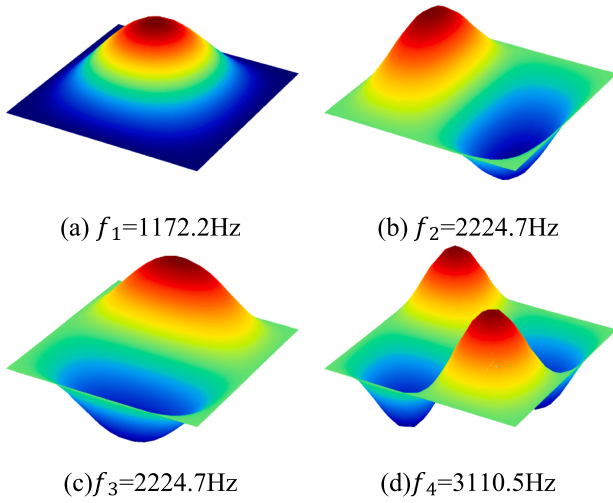


Fig. 8. Mode shapes of the plate without aerodynamic load.

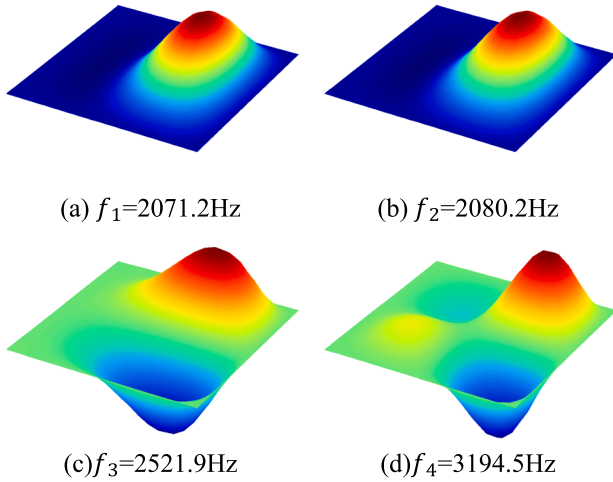


Fig. 9. Mode shapes of the plate when $\lambda = \lambda_{cr}$.

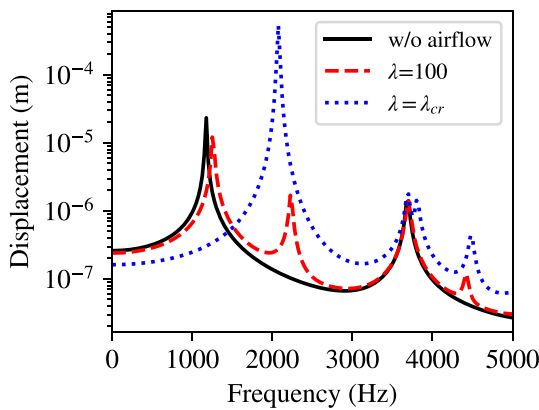


Fig. 10. RMS response of the plate with different aerodynamic pressure under acoustic load.

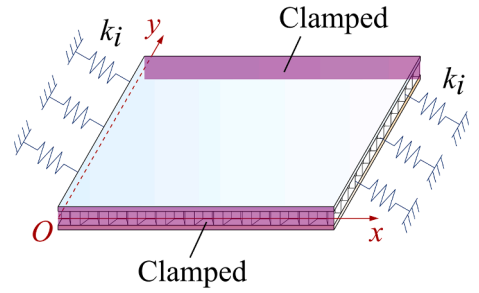


Fig. 11. Boundary conditions of the honeycomb plate.

study in the previous subsection.

3.2.1. Stability and forced response

The aeroelastic characteristics of the plate under aerodynamic load is studied first by solving its eigenvalue problem. It is known that the plate is stable when the real parts of all its eigenvalues are negative, and the plate becomes unstable when positive real part appears. Fig. 7(a) shows the real part of the eigenvalues. From the figure, it can be observed that the real parts of the eigenvalues are negative and almost constant and when λ is small. However, as λ keeps increasing and reaches the non-dimensional critical flutter aerodynamic pressure (CFAP) $\lambda_{cr} = 308.5$, the real parts of its first two eigenvalues split into two branches, one of which goes down and the other one quickly increases and becomes positive, which implies the system becomes unstable. Fig. 7(b) shows the imaginary part of the eigenvalues, which is in proportion to its natural frequency by a factor of 2π . The figure shows that when $\lambda < \lambda_{cr}$, the first natural frequency increases while the second natural frequency decreases, and they would coalesce near λ_{cr} . Thus, finding the intersection point of the natural frequencies provides another way for determining the flutter boundaries. The mode shapes of the plate without aerodynamic load and at CFAP are respectively shown in Fig. 8 and Fig. 9. It is obvious that the mode shapes change significantly when aerodynamic load is applied to the plate. The symmetry of the second and third mode shape of the plate, which is a result of the isotropy of the material property of the honeycomb layer, is also broken by the aerodynamic load. Another interesting finding is that the first two modes coincide when the aerodynamic pressure is equal to CFAP, and these two modes evolve from the first two modes of the plate without aerodynamic pressure. It should be kept in mind that although these two modes share the same mode shape, the stabilities of these two modes are different as explained above.

The response of the plate under acoustic load is also studied. The RMS response of the plate under unit acoustic load is shown in Fig. 10. It should be mentioned that in this study, an additional structural damping $\eta = 0.01$ is applied to the plate to ensure finite response. From the figure, it is obvious that the maximum response of the plate at CFAP is much larger than that without airflow. Another finding is that the symmetric modes do not contribute to the forced response of the plate at first, but they would appear when aerodynamic pressure is applied to the plate. This is because aerodynamic load would break the symmetry of the modes as discussed before.

3.2.2. Boundary conditions

In this study, numerical studies on different boundaries conditions are carried out. As shown in Fig. 11, the plate has two opposite edges clamped and the one of the five DOFs of the other two edges are constrained by a pair of elastic supports k_i , which is defined in Eq. (20). The nature frequencies of the plate with different k_i ($i = 1, 2, 3, 4, 5$) are respectively shown in Fig. 12. Fig. 12 (a) and Fig. 12 (b) show that the variation of in-plane translational stiffness k_1 and k_2 do not affect the natural frequencies of the plate. This is because the in-plane deformation of such plates only occurs at very high frequencies. Thus, the in-plane

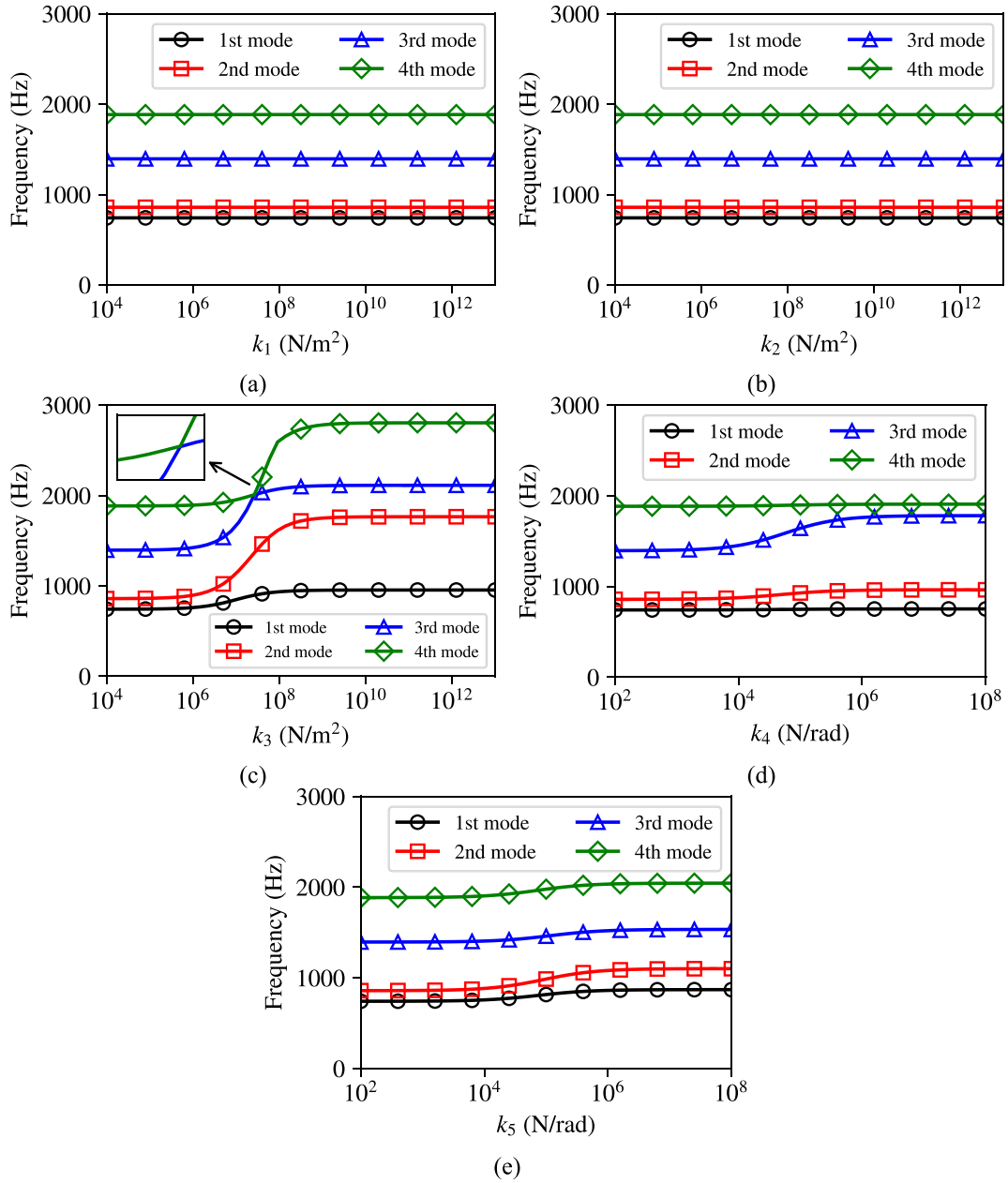


Fig. 12. Natural frequencies of the honeycomb plate with different boundary stiffness.

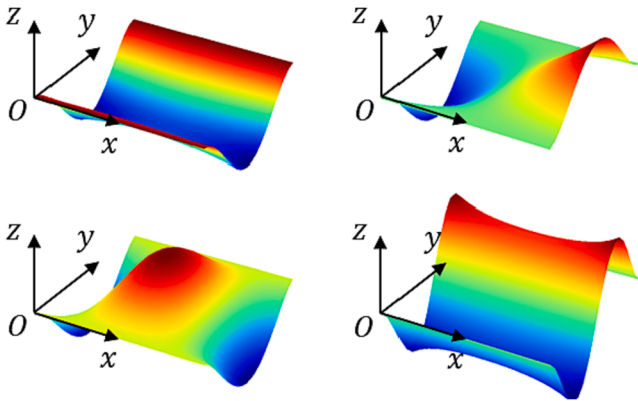


Fig. 13. Mode shapes of the honeycomb plate with C-F-C-F boundary conditions.

stiffness has little effect on the lower modes of the plate. However, as can be seen from Fig. 12 (c), when the out-of-plane translational stiffness k_3 increases, the natural frequencies of the plate go up rapidly as expected. Similar behaviors can be observed that with the increase of k_4 and k_5 , the natural frequencies of the plate also tend to increase. However, it can be found from Fig. 12(d) that the first and fourth natural frequencies remain constant when k_4 increases. To explain this behavior, the mode shapes of the plate should be studied first. Fig. 13 shows the first four mode shapes of the plate with C-F-C-F boundaries, which is just the case when $k_4 = 0$. It can be observed that for the first and fourth mode shape, the normal of the plate on the two free edges does not rotate about the y-axis, which means ϕ_x always equals to zero and increasing of k_4 will not contribute to the energy functional given by Eq. (20). Thus, modifying k_4 has no effect on these two modes.

3.2.3. Layer thickness

Layer thickness is one of the key parameters for sandwich honeycomb plates. In practical engineering, the total thickness of the plate is

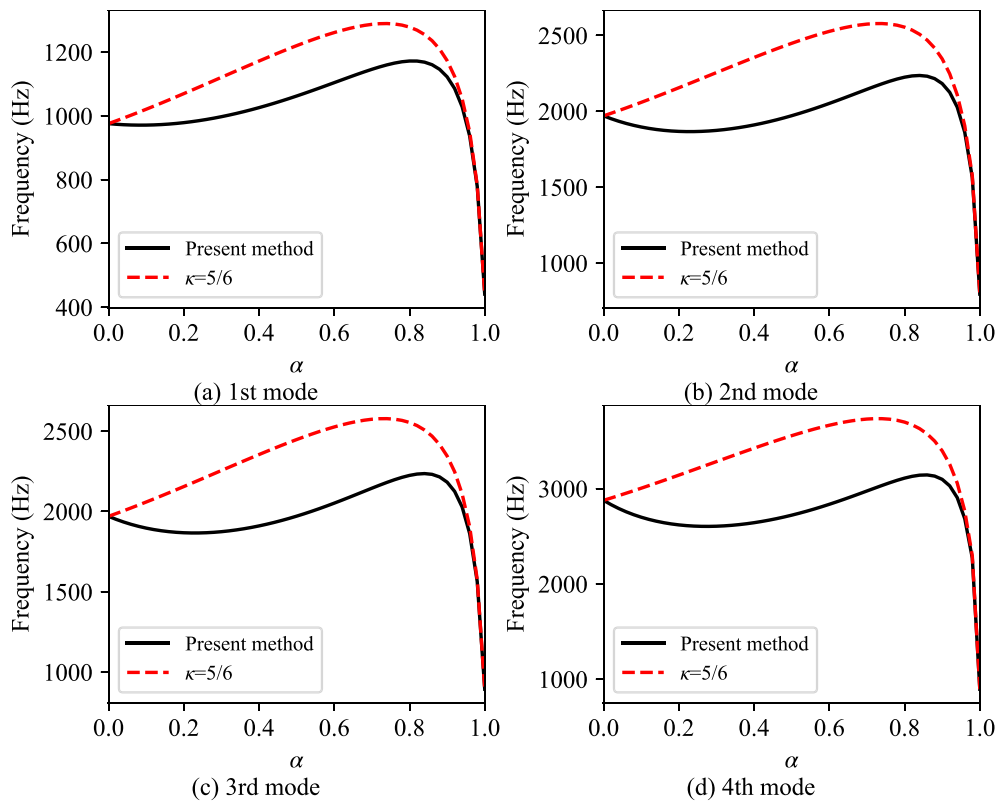


Fig. 14. Natural frequencies of the plate with different thickness ratio.

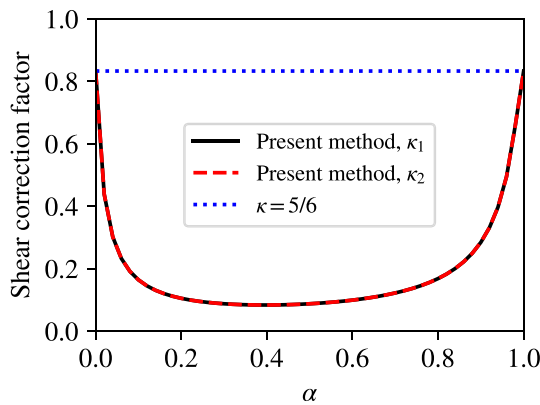


Fig. 15. Shear correction factor with different thickness ratio.

usually strictly restricted, and only the thickness ratio of the honeycomb core and two face sheets can be adjusted. Thus, in this study, the thicknesses of two face layers are assumed to be equal and the total thickness h of the plate is kept constant. By defining thickness ratio $\alpha = h_2/h$, the thicknesses of the two face layers can be represented as $h_1 = h_3 = (1-\alpha)h/2$. The effect of thickness ratio on the first four natural frequencies of the plate is shown in Fig. 14. From the figure, it can be found that when α increases, the natural frequencies decrease first and then increase when α is not very large. However, when α is close to 1, the natural frequencies of the plate decrease rapidly. It is worthwhile to mention that in this study, the shear correction factor is defined by Eqs. (10)–(12), which is different from the conventional definition $\kappa = 5/6$. Fig. 14 compares the natural frequencies predicted by different shear correction factors. The results show that letting $\kappa = 5/6$ leads to a large error when $0 < \alpha < 0.9$. This is due to the fact that directly using $\kappa = 5/6$ significantly overestimates the shear effect as shown in Fig. 15, which

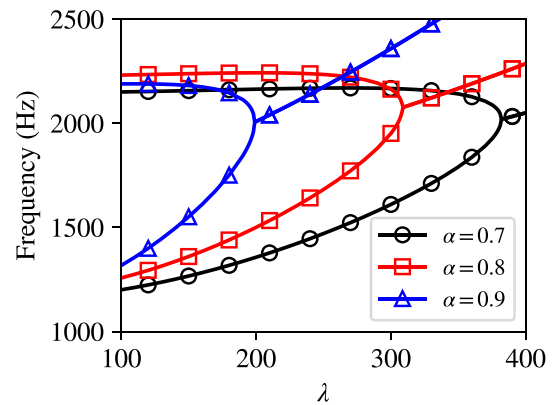


Fig. 16. First two natural frequencies of the plate with different thickness ratio under aerodynamic load.

stiffens the structure and results in a higher natural frequency. Fig. 16 shows the variations of the natural frequencies of the plate with different thickness ratio α . From the figure, it is clear that with the increase of the thickness ratio, the CFAP decreases.

The forced response of the plate with different thickness ratio is studied as well. The RMS response of the plate under unit acoustic excitation without aerodynamic pressure is shown in Fig. 17(a). The figure shows that the response of the plate becomes larger with the increase of the thickness ratio. Fig. 17(b) shows the response of the plate at CFAP. Comparing Fig. 17(b) with Fig. 17(a), it can be found that the frequencies of the peaks change and the maximum response becomes much larger when the aerodynamic pressure of the plate is close to CFAP. Meanwhile, the figure shows that although the CFAPs are quite different, the responses of the plate under CFAPs are close to each other for the three different thickness ratios.

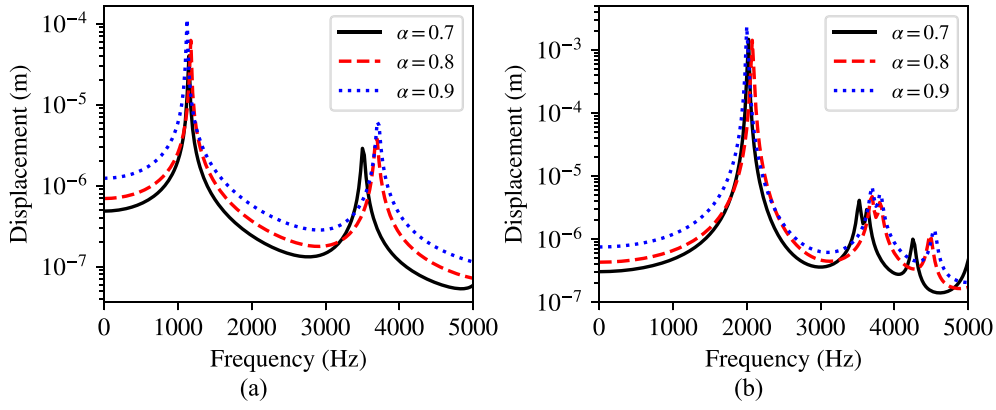


Fig. 17. RMS response of the plate with different thickness ratio: (a) without aerodynamic pressure, (b) at CFAP.

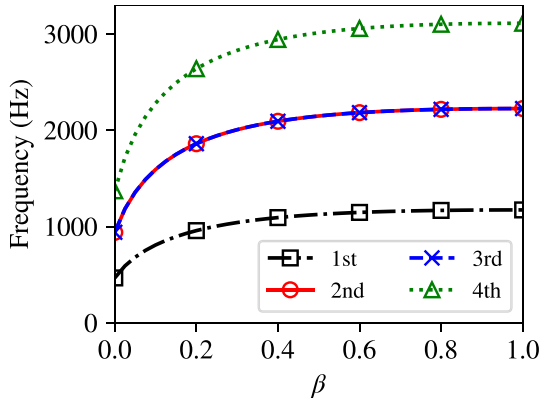


Fig. 18. First four natural frequencies of the plate with different β .

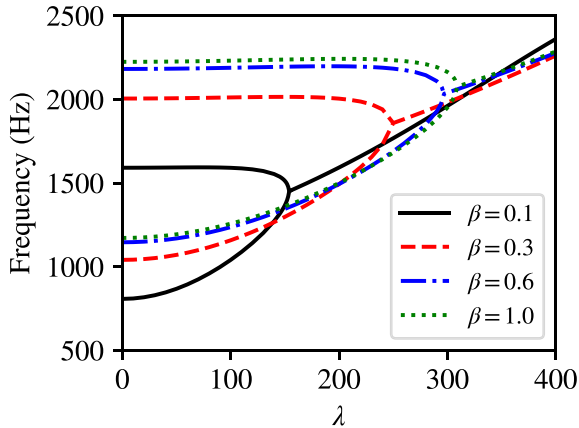


Fig. 19. First two natural frequencies of the plate with different β under aerodynamic load.

In the previous study, the layers of the plate are symmetric as the thickness of the two face layers are the same. In this study, however, the vibration of unsymmetric layers will be investigated. Letting $\beta = h_1/h_3$, and the thickness of the core and the total thickness of the core remain constant. The natural frequencies of the plate with different β are calculated and shown in Fig. 18. Obviously, the natural frequencies of the plate raise monotonously and achieves its maximum value when the two face layers are of equal thickness. Fig. 19 shows the variations of the natural frequencies of the plate with four different values of β . The result indicates that λ_{cr} quickly goes up when β increases and converges to its

maximum value when β equals 1, that is, CFAP reaches its maximum value when the thickness values of two face layers of the plate are identical.

3.2.4. Honeycomb geometry

The influence of the honeycomb parameters on the dynamics of the plate is investigated. From Eq. (8), it can be found that the equivalent material parameters of the honeycomb core depend on the non-dimensional parameters η_1 , η_2 and the cell angle θ_c . The influence of these parameters on the natural frequencies of the plate is shown in Fig. 20. It can be seen from the figure that with the increase of η_1 , the natural frequencies of the plate quickly increase at first. But when the value of η_1 keeps increasing, the natural frequencies would go down. Another phenomenon is that when the value of η_1 is far away from 1, the natural frequencies of the second and third modes are no longer coincident. Since $\eta_1 = l_{2c}/l_{1c}$ denotes the ratio of hexagon edge length, the results show that changing the shape of the hexagons would break the isotropy of the honeycomb. The effect of non-dimensional cell wall thickness η_2 on the free vibration of the plate is given in Fig. 20(b). Similarly, the natural frequencies of the plate firstly go up and then decrease. The natural frequencies of a structure depend on its mass and stiffness, larger wall thickness would increase the stiffness of the core layer, which leads to a higher natural frequency. However, when the wall thickness keeps increasing, more extra mass is added to the plate and finally lowers the natural frequencies of the plate. The influence of cell angle θ_c is also studied and shown in Fig. 20(c). From the figure, the natural frequencies of the honeycomb plate remain almost constant when θ_c is not vary large. However, the natural frequencies drop quickly after θ_c reaches 60° .

To further study the effect of the honeycomb geometry parameters on the flutter behaviors of the plate, the variations of the CFAP are calculated and shown in Fig. 21. It can be seen from the figure that despite the changes of the natural frequencies, CFAP decreases with η_1 and increases with η_2 monotonously. And with the increase of cell angle θ_c , CFAP decreases when θ_c is smaller than 10° or larger than 75° , and increases in other circumstances.

3.2.5. Honeycomb orientation angle

The effect of the honeycomb orientation angle θ_2 on the natural frequencies of the plate is investigated. From Fig. 22, it could be found that the natural frequencies of the plate remain constant when $\theta_c = 30^\circ$. This is due to the fact that the equivalent material properties of the core layer are almost isotropic when $\theta_c = 30^\circ$. However, when $\theta_c = 0^\circ$ or 60° , the material becomes anisotropic and the natural frequencies of the plate begin to change with θ_2 . From the figures, it is obvious that the natural frequencies of the plate achieve its maximum value when $\theta_2 = 45^\circ$. But in these two cases, even though at $\theta_2 = 45^\circ$, their natural frequencies are lower than those of the case when $\theta_c = 30^\circ$. Another finding

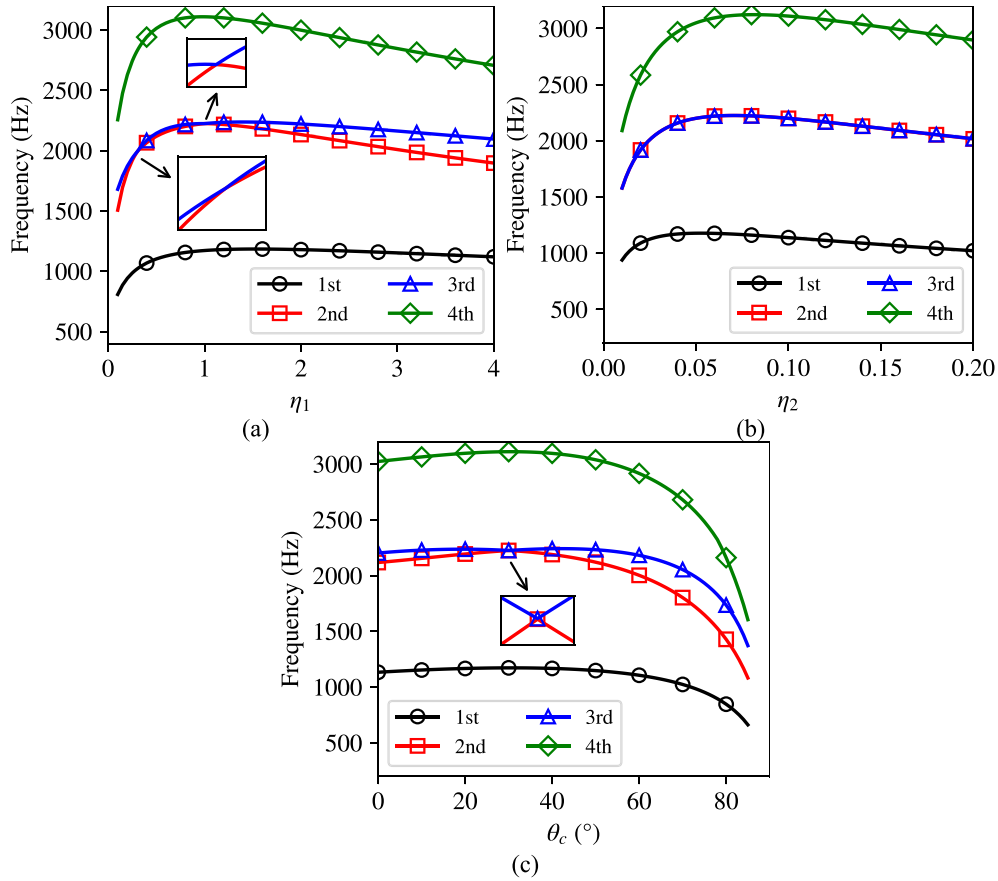


Fig. 20. Variations of the first four natural frequencies with different honeycomb parameters: (a) aspect ratio η_1 , (b) non-dimensional cell wall thickness η_2 , (c) cell angle θ_c .

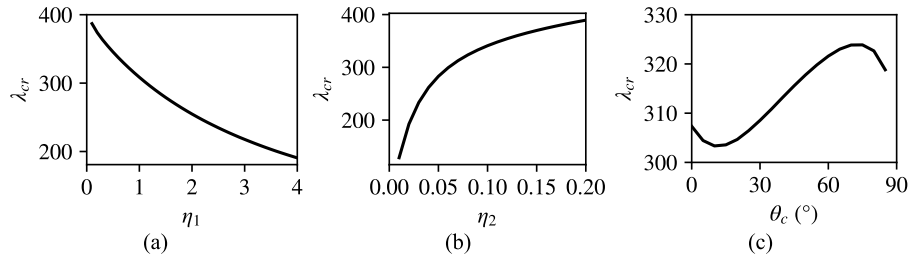


Fig. 21. Variations of the CFAP with different honeycomb parameters: (a) aspect ratio η_1 , (b) non-dimensional cell wall thickness η_2 , (c) cell angle θ_c .

is that the natural frequencies are symmetric about $\theta_2 = 45^\circ$, as the plate itself is symmetric about the x - and y - axis. To deeper look into the phenomenon, the variations of the natural frequencies of the plate with non-dimensional aerodynamic pressure $\lambda = 200$ are shown in Fig. 23. From the figure, although the natural frequencies at $\theta_c = 30^\circ$ still remain constant due to its isotropic material properties, the other curves are no longer symmetric, which shows the natural frequencies of the plate is sensitive to θ_2 when θ_c does not equal to 30° . The variations of the CFAP with θ_2 for the three cases are also calculated and shown in Fig. 24. As explained above, the CFAP is not sensitive to θ_2 when $\theta_c = 30^\circ$ due to its isotropy, but changes significantly when θ_c equals 0° or 60° .

4. Conclusion

In this paper, the modeling and aeroelastic properties of laminated plates with honeycomb core is investigated. By adopting the corrected Gibson's formula, the honeycomb layer is modeled by its equivalent material properties. Based on the FSDT, the energy functionals of the

plate with general boundary conditions and acoustic load are formulated, and the aerodynamic pressure is taken into consideration by using the supersonic piston theory. To ensure accuracy for plates with soft cores, a modified shear correction factor is adopted in this study to accurately model the out-of-plane shear stresses of the plate. The Hamilton's principle is employed to formulate the governing equations and the equations are solved by a modified Ritz approach.

To show the reliability of the proposed method, a computer program is developed and the results are compared with those from FEM and other studies. A good agreement is observed from the results and the developed computer program shows to be much more efficient than the FEM. Finally, the influences of some key parameters of sandwich honeycomb plates on the vibration and aeroelastic characteristics are analyzed. It is found that for sandwich plates with soft core, modified shear correction factor should be used to get accurate results if the plate has moderate thick face layers. Moreover, numerical studies show that proper modification of the honeycomb cell geometry may break the symmetry of the equivalent material properties and leads to changes of

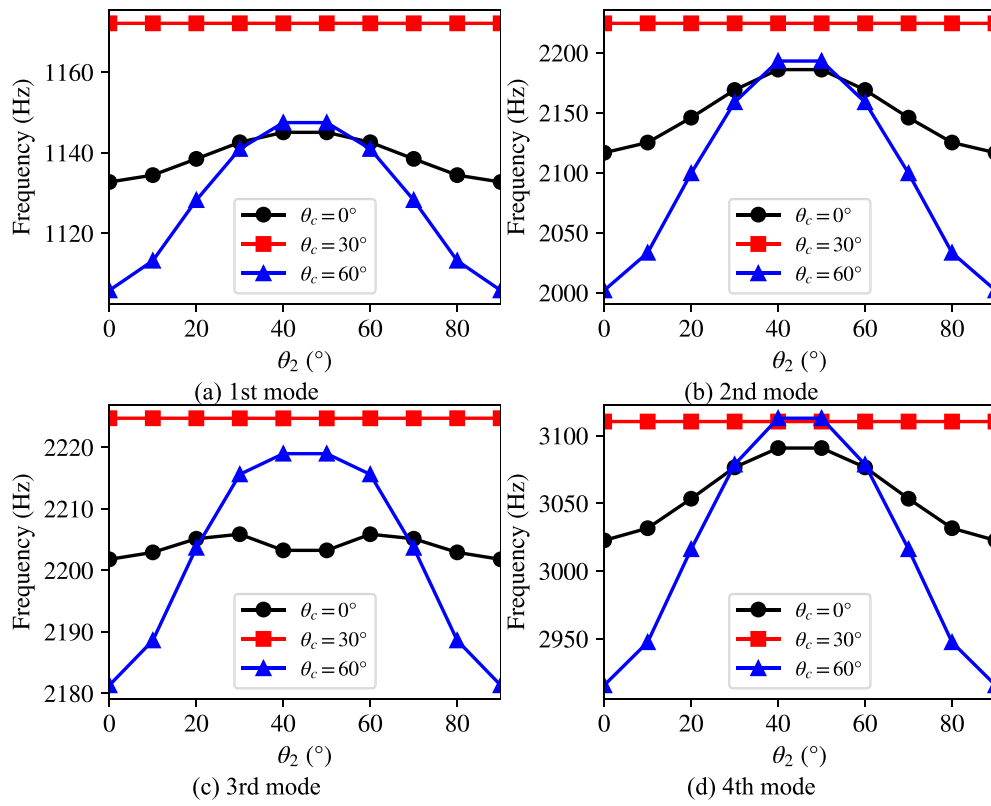


Fig. 22. Variations of the first four natural frequencies with different θ_2 without airflow.

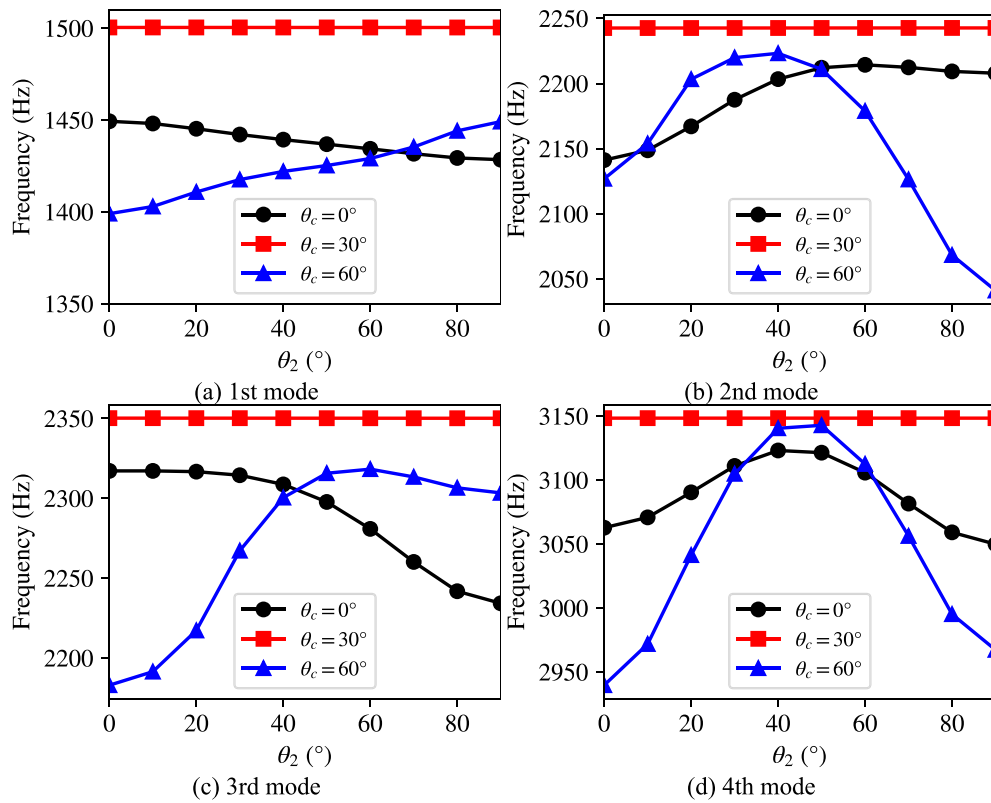


Fig. 23. Variations of the first four natural frequencies with different θ_2 with $\lambda = 200$.

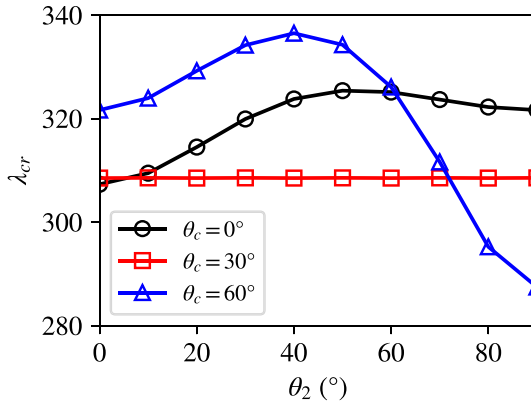


Fig. 24. Variations of the CFAP with different θ_2 .

the aeroelastic behaviors of the whole structure.

5. Data availability statement

The data in this study will be available on request from the corresponding author.

Appendix A

The governing equations of the plate can be obtained from Eq. (26) by setting the coefficients of δu_0 , δv_0 , δw_0 , $\delta \phi_x$ and $\delta \phi_y$ to zero separately. The governing equations of a plate with four edges elastically supported are

$$A_{11} \frac{\partial^2 u_0}{\partial x^2} + A_{12} \frac{\partial^2 v_0}{\partial x \partial y} + A_{16} \left(2 \frac{\partial^2 u_0}{\partial x \partial y} + \frac{\partial^2 v_0}{\partial x^2} \right) + A_{26} \frac{\partial^2 v_0}{\partial y^2} + A_{66} \left(\frac{\partial^2 u_0}{\partial y^2} + \frac{\partial^2 v_0}{\partial x \partial y} \right) + B_{11} \frac{\partial^2 \phi_x}{\partial x^2} + B_{12} \frac{\partial^2 \phi_y}{\partial x \partial y} + B_{16} \left(2 \frac{\partial^2 \phi_x}{\partial x \partial y} + \frac{\partial^2 \phi_y}{\partial x^2} \right) + B_{26} \frac{\partial^2 \phi_y}{\partial y^2} + B_{66} \left(\frac{\partial^2 \phi_x}{\partial y^2} + \frac{\partial^2 \phi_y}{\partial x \partial y} \right) = I_0 \ddot{u}_0 + I_1 \ddot{\phi}_x \quad (\text{A-1})$$

$$A_{12} \frac{\partial^2 u_0}{\partial y \partial x} + A_{16} \frac{\partial^2 u_0}{\partial x^2} + A_{26} \left(\frac{\partial^2 u_0}{\partial y^2} + \frac{\partial^2 v_0}{\partial x \partial y} \right) + A_{66} \left(\frac{\partial^2 u_0}{\partial x \partial y} + \frac{\partial^2 v_0}{\partial x^2} \right) + A_{22} \frac{\partial^2 v_0}{\partial y^2} + B_{12} \frac{\partial^2 \phi_x}{\partial x \partial y} + B_{16} \frac{\partial^2 \phi_x}{\partial x^2} + B_{22} \frac{\partial^2 \phi_y}{\partial y^2} + B_{26} \left(\frac{\partial^2 \phi_x}{\partial y^2} + 2 \frac{\partial^2 \phi_y}{\partial x \partial y} \right) + B_{66} \left(\frac{\partial^2 \phi_y}{\partial x^2} + \frac{\partial^2 \phi_x}{\partial x \partial y} \right) = I_0 \ddot{v}_0 + I_1 \ddot{\phi}_y \quad (\text{A-2})$$

$$A_{44} \left(\kappa_1 \frac{\partial^2 w_0}{\partial y^2} + \kappa_1 \frac{\partial \phi_y}{\partial y} \right) + A_{45} \left(\kappa_1 \frac{\partial \phi_x}{\partial y} + \kappa_1 \frac{\partial^2 w_0}{\partial x \partial y} + \kappa_2 \frac{\partial^2 w_0}{\partial x \partial y} + \kappa_2 \frac{\partial \phi_y}{\partial x} \right) + A_{55} \left(\kappa_2 \frac{\partial^2 w_0}{\partial x^2} + \kappa_2 \frac{\partial \phi_x}{\partial x} \right) + \frac{\rho_\infty U_\infty^2}{\sqrt{M_\infty^2 - 1}} \left(\frac{M_\infty^2 - 2}{M_\infty^2 - 1} \frac{1}{U_\infty} \dot{w} + \frac{\partial w}{\partial x} \cos \theta_a + \frac{\partial w}{\partial y} \sin \theta_a \right) + p_s = I_0 \ddot{w}_0 \quad (\text{A-3})$$

$$-A_{45} \left(\kappa_2 \phi_y + \kappa_2 \frac{\partial w_0}{\partial y} \right) - A_{55} \left(\kappa_2 \phi_x + \kappa_2 \frac{\partial w_0}{\partial x} \right) + B_{11} \frac{\partial^2 u_0}{\partial x^2} + B_{12} \frac{\partial^2 v_0}{\partial x \partial y} + B_{16} \left(\frac{\partial^2 v_0}{\partial x^2} + 2 \frac{\partial^2 u_0}{\partial x \partial y} \right) + B_{26} \frac{\partial^2 v_0}{\partial y^2} + B_{66} \left(\frac{\partial^2 u_0}{\partial y^2} + \frac{\partial^2 v_0}{\partial x \partial y} \right) + D_{11} \frac{\partial^2 \phi_x}{\partial x^2} + D_{12} \frac{\partial^2 \phi_y}{\partial x \partial y} + D_{16} \left(\frac{\partial^2 \phi_y}{\partial x^2} + 2 \frac{\partial^2 \phi_x}{\partial x \partial y} \right) + D_{26} \frac{\partial^2 \phi_y}{\partial y^2} + D_{66} \left(\frac{\partial^2 \phi_x}{\partial y^2} + \frac{\partial^2 \phi_y}{\partial x \partial y} \right) = I_1 \ddot{u}_0 + I_2 \ddot{\phi}_x \quad (\text{A-4})$$

$$-A_{44} \left(\kappa_1 \phi_y + \kappa_1 \frac{\partial w_0}{\partial y} \right) - A_{45} \left(\kappa_1 \phi_x + \kappa_1 \frac{\partial w_0}{\partial x} \right) + B_{12} \frac{\partial^2 u_0}{\partial x \partial y} + B_{16} \frac{\partial^2 u_0}{\partial x^2} + B_{22} \frac{\partial^2 v_0}{\partial y^2} + B_{26} \left(\frac{\partial^2 u_0}{\partial y^2} + 2 \frac{\partial^2 v_0}{\partial y \partial x} \right) + B_{66} \left(\frac{\partial^2 v_0}{\partial x^2} + \frac{\partial^2 u_0}{\partial x \partial y} \right) + D_{12} \frac{\partial^2 \phi_x}{\partial x \partial y} + D_{16} \frac{\partial^2 \phi_x}{\partial x^2} + D_{22} \frac{\partial^2 \phi_y}{\partial y^2} + D_{26} \left(\frac{\partial^2 \phi_x}{\partial y^2} + 2 \frac{\partial^2 \phi_y}{\partial x \partial y} \right) + D_{66} \left(\frac{\partial^2 \phi_y}{\partial x^2} + \frac{\partial^2 \phi_x}{\partial x \partial y} \right) = I_1 \ddot{v}_0 + I_2 \ddot{\phi}_y \quad (\text{A-5})$$

The corresponding boundary conditions along the x-axis are

$$A_{16} \frac{\partial u_0}{\partial x} + A_{26} \frac{\partial v_0}{\partial y} + A_{66} \left(\frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x} \right) + B_{16} \frac{\partial \phi_x}{\partial x} + B_{26} \frac{\partial \phi_y}{\partial y} + B_{66} \left(\frac{\partial \phi_x}{\partial y} + \frac{\partial \phi_y}{\partial x} \right) + k_1 u_0 = 0 \quad (\text{A-6})$$

$$A_{12} \frac{\partial u_0}{\partial x} + A_{22} \frac{\partial v_0}{\partial y} + A_{26} \left(\frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x} \right) + B_{12} \frac{\partial \phi_x}{\partial x} + B_{22} \frac{\partial \phi_y}{\partial y} + B_{26} \left(\frac{\partial \phi_x}{\partial y} + \frac{\partial \phi_y}{\partial x} \right) + k_2 v_0 = 0 \quad (\text{A-7})$$

$$A_{44} \left(\kappa_1 \phi_y + \kappa_1 \frac{\partial w_0}{\partial y} \right) + A_{45} \left(\kappa_1 \phi_x + \kappa_1 \frac{\partial w_0}{\partial x} \right) + k_3 w_0 = 0 \quad (\text{A-8})$$

CRediT authorship contribution statement

Zhen Ni: Conceptualization, Software, Formal analysis, Writing – original draft. **Danwang Li:** Data curation, Validation, Resources. **Liang Ji:** Supervision, Project administration, Funding acquisition. **Kai Zhou:** Methodology, Investigation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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$$B_{16}\frac{\partial u_0}{\partial x} + B_{26}\frac{\partial v_0}{\partial y} + B_{66}\left(\frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x}\right) + D_{16}\frac{\partial \phi_x}{\partial x} + D_{26}\frac{\partial \phi_y}{\partial y} + D_{66}\left(\frac{\partial \phi_x}{\partial y} + \frac{\partial \phi_y}{\partial x}\right) + k_4\phi_x = 0 \quad (\text{A-9})$$

$$B_{12}\frac{\partial u_0}{\partial x} + B_{22}\frac{\partial v_0}{\partial y} + B_{26}\left(\frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x}\right) + D_{12}\frac{\partial \phi_x}{\partial x} + D_{22}\frac{\partial \phi_y}{\partial y} + D_{26}\left(\frac{\partial \phi_x}{\partial y} + \frac{\partial \phi_y}{\partial x}\right) + k_5\phi_y = 0 \quad (\text{A-10})$$

And the boundary conditions along the y-axis are

$$A_{11}\frac{\partial u_0}{\partial x} + A_{12}\frac{\partial v_0}{\partial y} + A_{16}\left(\frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x}\right) + B_{11}\frac{\partial \phi_x}{\partial x} + B_{12}\frac{\partial \phi_y}{\partial y} + B_{16}\left(\frac{\partial \phi_x}{\partial y} + \frac{\partial \phi_y}{\partial x}\right) + k_1u_0 = 0 \quad (\text{A-11})$$

$$A_{16}\frac{\partial u_0}{\partial x} + A_{26}\frac{\partial v_0}{\partial y} + A_{66}\left(\frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x}\right) + B_{16}\frac{\partial \phi_x}{\partial x} + B_{26}\frac{\partial \phi_y}{\partial y} + B_{66}\left(\frac{\partial \phi_x}{\partial y} + \frac{\partial \phi_y}{\partial x}\right) + k_2v_0 = 0 \quad (\text{A-12})$$

$$A_{45}\left(\kappa_2\phi_y + \kappa_2\frac{\partial w_0}{\partial y}\right) + A_{55}\left(\kappa_2\phi_x + \kappa_2\frac{\partial w_0}{\partial x}\right) + k_3w_0 = 0 \quad (\text{A-13})$$

$$B_{11}\frac{\partial u_0}{\partial x} + B_{12}\frac{\partial v_0}{\partial y} + B_{16}\left(\frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x}\right) + D_{11}\frac{\partial \phi_x}{\partial x} + D_{12}\frac{\partial \phi_y}{\partial y} + D_{16}\left(\frac{\partial \phi_x}{\partial y} + \frac{\partial \phi_y}{\partial x}\right) + k_4\phi_x = 0 \quad (\text{A-14})$$

$$B_{16}\frac{\partial u_0}{\partial x} + B_{26}\frac{\partial v_0}{\partial y} + B_{66}\left(\frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x}\right) + D_{16}\frac{\partial \phi_x}{\partial x} + D_{26}\frac{\partial \phi_y}{\partial y} + D_{66}\left(\frac{\partial \phi_x}{\partial y} + \frac{\partial \phi_y}{\partial x}\right) + k_5\phi_y = 0 \quad (\text{A-15})$$

Appendix B

The detailed expression of the mass matrix $\mathbf{M}_s$ for the plate is

$$\mathbf{M}_s = \begin{pmatrix} \mathbf{M}_0 & 0 & 0 & \mathbf{M}_1 & 0 \\ 0 & \mathbf{M}_0 & 0 & 0 & \mathbf{M}_1 \\ 0 & 0 & \mathbf{M}_0 & 0 & 0 \\ \mathbf{M}_1 & 0 & 0 & \mathbf{M}_2 & 0 \\ 0 & \mathbf{M}_1 & 0 & 0 & \mathbf{M}_2 \end{pmatrix} \quad (\text{B-1})$$

where

$$\mathbf{M}_i = \int_{\Omega} I_i \Psi \Psi^T d\Omega \quad (\text{B-2})$$

The detailed expression of the stiffness matrix $\mathbf{K}_s$ for the plate is

$$\mathbf{K}_s = \begin{pmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} & 0 & \mathbf{K}_{14} & \mathbf{K}_{15} \\ \mathbf{K}_{12}^T & \mathbf{K}_{22} & 0 & \mathbf{K}_{24} & \mathbf{K}_{25} \\ 0 & 0 & \mathbf{K}_{33} & \mathbf{K}_{34} & \mathbf{K}_{35} \\ \mathbf{K}_{14}^T & \mathbf{K}_{24}^T & \mathbf{K}_{34}^T & \mathbf{K}_{44} & \mathbf{K}_{45} \\ \mathbf{K}_{15}^T & \mathbf{K}_{25}^T & \mathbf{K}_{35}^T & \mathbf{K}_{45}^T & \mathbf{K}_{55} \end{pmatrix} \quad (\text{B-3})$$

where

$$\mathbf{K}_{ij} = \int_{\Omega} \left(\Psi \quad \frac{\partial \Psi}{\partial x} \quad \frac{\partial \Psi}{\partial y} \right) \mathbf{R}_{ij} \begin{pmatrix} \Psi^T \\ \frac{\partial \Psi^T}{\partial x} \\ \frac{\partial \Psi^T}{\partial y} \end{pmatrix} d\Omega \quad (\text{B-4})$$

and the expressions for $\mathbf{R}_{ij}$ are

$$\mathbf{R}_{11} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & A_{11} & A_{16} \\ 0 & A_{16} & A_{66} \end{pmatrix}, \mathbf{R}_{12} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & A_{16} & A_{12} \\ 0 & A_{66} & A_{26} \end{pmatrix}, \mathbf{R}_{14} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & B_{11} & B_{16} \\ 0 & B_{16} & B_{66} \end{pmatrix},$$

$$\mathbf{R}_{15} = \mathbf{R}_{24}^T = \begin{pmatrix} 0 & 0 & 0 \\ 0 & B_{16} & B_{12} \\ 0 & B_{66} & B_{26} \end{pmatrix}, \mathbf{R}_{22} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & A_{66} & A_{26} \\ 0 & A_{26} & A_{22} \end{pmatrix}, \mathbf{R}_{25} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & B_{66} & B_{26} \\ 0 & B_{26} & B_{22} \end{pmatrix},$$

$$\mathbf{R}_{33} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & A_{55} & A_{45} \\ 0 & A_{45} & A_{44} \end{pmatrix}, \mathbf{R}_{34} = \begin{pmatrix} 0 & 0 & 0 \\ A_{55} & 0 & 0 \\ A_{45} & 0 & 0 \end{pmatrix}, \mathbf{R}_{35} = \begin{pmatrix} 0 & 0 & 0 \\ A_{45} & 0 & 0 \\ A_{44} & 0 & 0 \end{pmatrix}$$

$$\mathbf{R}_{44} = \begin{pmatrix} A_{55} & 0 & 0 \\ 0 & D_{11} & D_{16} \\ 0 & D_{16} & D_{66} \end{pmatrix}, \mathbf{R}_{45} = \begin{pmatrix} A_{45} & 0 & 0 \\ 0 & D_{16} & D_{12} \\ 0 & D_{66} & D_{26} \end{pmatrix}, \mathbf{R}_{55} = \begin{pmatrix} A_{44} & 0 & 0 \\ 0 & D_{66} & D_{26} \\ 0 & D_{26} & D_{22} \end{pmatrix} \quad (\text{B-5})$$

The expression of the stiffness matrix $\mathbf{K}_{\text{BC}}$ for the boundary conditions is

$$\mathbf{K}_{\text{BC}} = \int_{\Omega} \Psi \mathbf{D}_k \Psi^T d\Omega \quad (\text{B-6})$$

where $\mathbf{D}_k = \text{diag}(\mathbf{K}_1, \mathbf{K}_2, \mathbf{K}_3, \mathbf{K}_4, \mathbf{K}_5)$ and $\mathbf{K}_i = k_i \mathbf{I}_{MN}$ and $\mathbf{I}_{MN}$ is the identity matrix with size $MN \times MN$.

The expression of the aerodynamic stiffness matrix $\mathbf{K}_a$ is

$$\mathbf{K}_a = \text{diag} \left(0, 0, \frac{\rho_{\infty} U_{\infty}^2}{\sqrt{M_{\infty}^2 - 1}} \left(\cos \theta_a \Psi \frac{\partial \Psi^T}{\partial x} + \sin \theta_a \Psi \frac{\partial \Psi^T}{\partial y} \right), 0, 0 \right) \quad (\text{B-7})$$

The expression of the aerodynamic damping matrix $\mathbf{C}_a$ is

$$\mathbf{C}_a = \text{diag} \left(0, 0, \frac{\rho_{\infty} U_{\infty}^2}{\sqrt{M_{\infty}^2 - 1}} \frac{M_{\infty}^2 - 2}{M_{\infty}^2 - 1} \frac{1}{U_{\infty}} \Psi \Psi^T, 0, 0 \right) \quad (\text{B-8})$$

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